Preserving lesion quantification in longitudinal whole-body PSMA PET/CT registration

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Abstract. Purpose: Registering follow-up to baseline PSMA PET/CT scans enables voxel-wise comparison of tumor burden, but the deformation can itself change the metabolic tumor volume (MTV) and total lesion glycolysis (TLG) of lesions. We examine how PET-preservation and bone-rigidity losses trade alignment for less registration-induced change, and whether they belong in network training or instance optimization (IO).

Approach: We trained LapIRN networks with and without the losses on the Learn2Reg 2026 PSMAReg data, applied them in training, IO, or both, compared three rigidity penalties, and evaluated on 28 held-out patients against tuned NiftyReg and ConvexAdam, with patient-level statistics and per-lesion analyses.

Results: With the losses only in IO, our method reached 78.3 % CT Dice (NiftyReg 70.1 %, ConvexAdam 61.7 %) with similar MTV and TLG change. [TODO: Baseline numbers after re-tuning.] IO without the losses increased the MTV and TLG change relative to network output; with the losses, IO gained 2.9 Dice points without a detected increase in either scan-level measure. Applying the losses only in IO gained 1.0 Dice point over applying them in both training and IO; whether preservation differs between these placements remains unresolved. Lesions missed by the predicted masks contributed more than half of the residual scan-level change in the PET-loss IO analysis. With provided masks, the scan-level MTV change fell to 0.1 %, but individual lesions still changed by a median of 13.5 %. A per-bone rigid fit preserved within-bone distances best (distortion 13.4 % to 5.0 %). When added to the PET losses, it produced no detected further reduction in lesion change.

Conclusions: With predicted lesion masks, IO with PET-preservation and bone-rigidity losses improved anatomical alignment without a detected increase in scan-level MTV or TLG change relative to network output. This option requires time for per-pair refinement; when IO is omitted, training with the losses reduces registration-induced PET change. Missed lesions limit scan-level preservation, while substantial changes to individual lesions remain even with provided masks. Per-lesion evaluation is therefore needed to judge preservation for lesion-level applications.

Keywords: deformable image registration, PSMA PET/CT, longitudinal imaging, biomarker preservation, instance optimization.

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1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT is now routinely used for staging prostate cancer, for planning and following radioligand therapy, and for assessing response.[1] Longitudinal imaging provides a baseline examination and one or more follow-up examinations. Registering each follow-up to the baseline makes it possible to compare tumor burden voxel by voxel, track

individual lesions, and accumulate dose.[2] For these uses, a registration must do two things at once: it has to align the anatomy, and it must not itself change the PET-derived quantities that are compared across time points. Two such quantities are the metabolic tumor volume (MTV) and the total lesion glycolysis (TLG). A deformation that compresses or stretches a lesion changes both, so a change measured after registration can be an artifact of the registration rather than of the disease. The Learn2Reg 2026 PSMAReg challenge[3–5] formalizes this setting: each follow-up scan is warped to its baseline by a dense displacement field, which is scored on organ overlap, boundary distance, folding, and the change of MTV and TLG caused by the warp. The two goals can compete: limiting deformation may protect lesion measurements while leaving the anatomy misaligned, whereas improving alignment may distort the lesions being measured.

Classical registration solves an optimization problem per image pair;[6–8] this is accurate but slow for whole-body scans. Learning-based registration replaces the per-pair optimization with one forward pass of a trained network,[9–11] and instance optimization (IO) can refine the network output per pair at test time. The coarse-to-fine Laplacian pyramid network LapIRN[12] recovers the large deformations between whole-body scans while keeping the transformation smooth and invertible. With a network and optional IO, preservation constraints can act during training, during per-pair refinement, or at both stages; the choice affects both registration quality and runtime. The label-guided IO studied here uses predicted organ and lesion masks for each pair; we evaluate its output against the provided labels. Its PET-preservation losses act on the predicted lesion mask, so a lesion absent from that mask receives no direct preservation constraint.

Our earlier Learn2Reg submission introduced the affine–LapIRN–IO pipeline with PET-preservation and per-bone rigidity losses and evaluated its selected configurations on challenge validation cases using surrogate labels.[13] In this extension, we retrain on a new patient-level split of the released labeled cohort, reserve 28 patients for final evaluation, and add controlled loss-placement comparisons, tuned baselines, patient-level statistics, and per-lesion analyses. These 28 patients were among the challenge data available during development of the earlier method, so the split provides a holdout for the present analyses rather than an independent external cohort.

First, we ask whether the PET-preservation and bone-rigidity losses reduce registration-induced change and what alignment accuracy they cost, both with and without IO. We then ask whether applying the losses only during IO can recover the preservation obtained by also using them in training, and compare that choice with the faster network-only setting. Finally, we analyze changes to individual lesions and the coverage of the predicted lesion masks used by IO to understand what limits preservation. Because many PSMA lesions lie in bone, we also compare rigidity penalties for low-resolution whole-body CT, where thin skeletal structures make local constraints difficult to apply. Our contributions are:

- a controlled comparison of where PET-preservation and rigidity losses act (network training, IO, or both), showing that loss-guided IO improves alignment without a detected increase in scan-level MTV or TLG change relative to network output, and that applying the losses only in IO gives higher alignment accuracy than using them at both stages;
- an analysis of the remaining change, showing that missed lesions account for most of the residual scan-level PET change in the mask-coverage analysis, while scan-level MTV and TLG hide substantial changes to individual lesions even when provided masks are used;

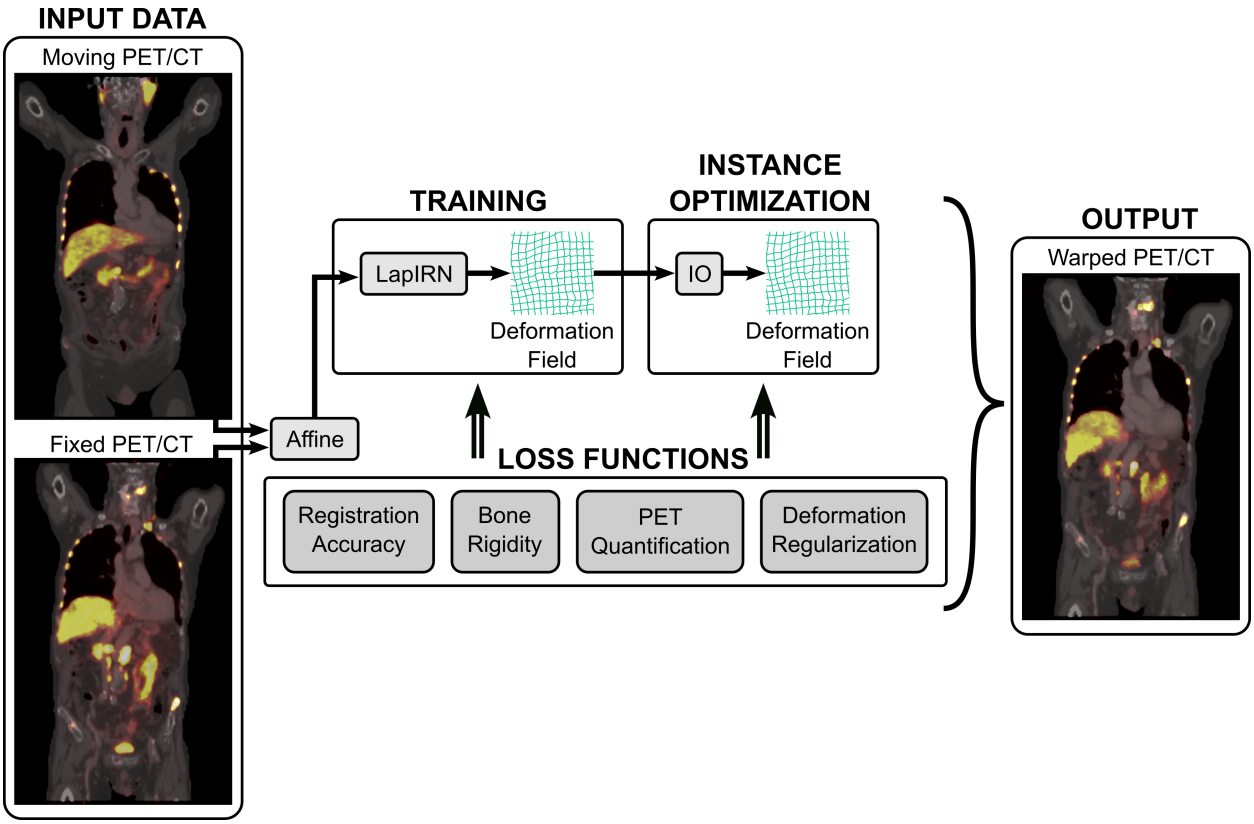


Fig 1 Registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT scans are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimization (IO) refines this deformation per pair before the moving scan is resampled into the fixed frame. Network training and IO use the same groups of loss terms: registration accuracy, bone rigidity, PET quantification, and deformation regularization (Sec. 2.3). The bone-rigidity and PET-quantification terms can be used in training, in IO, or in both; our method uses them only in IO.

- a comparison of rigidity penalties for whole-body CT, showing that a per-bone rigid fit avoids the boundary leakage of finite-difference penalties and the loss of skeleton under erosion, and preserves within-bone distances best;
- a patient-level evaluation with the official challenge scorer, a selection rule that weighs preservation against a fixed Dice tolerance, and tuned NiftyReg and ConvexAdam baselines.

2 Methods

2.1 Data and Split

We used the training set of the PSMAReg challenge,[2,5] a longitudinal whole-body PSMA PET/CT cohort. Each scan consists of a CT and a PET image (in SUV), resampled to $2.7344 \times 2.7344 \times 3.27$ mm and cropped to $192 \times 192 \times 288$ voxels. For every scan, the challenge provides 117 CT organ labels (the TotalSegmentator[14] classes) and a PSMA lesion mask. Following the challenge, a follow-up scan is always the moving image and the corresponding baseline scan the fixed image.

The original dataset publication reports that release of the anonymized retrospective imaging data was approved by the Ethics Committee of the Medical Faculty, LMU Munich (applications 21-1230 and 21-0098), and by the institutional data security and privacy review board.[15] It also reports that the ethics committee waived individual informed consent because the images were fully anonymized. The present study is a secondary analysis of publicly released challenge data and involved no patient recruitment or contact.

We split the 146 labeled patients with at least two scans at the patient level into training, validation, and test sets, aiming at 70/15/15 % of the registration pairs and balancing the sets for tumor burden and for bone- versus soft-tissue lesion location. Training uses all pairs of later onto earlier examinations of a patient; validation and testing use only follow-up-to-baseline pairs, as in the challenge. This gave 92/26/28 patients and 178/38/38 pairs. The split was built once with a fixed seed and frozen before any experiment of this study; all choices of settings, checkpoints, and methods were made on the validation set, and the test set was used only for the final evaluation. All patients were part of the challenge training data used to develop the earlier version of the method;[13] the test patients were not used for any decision in the present study.

Labels used by the method and for evaluation. Network training uses the provided organ and lesion labels of the training patients. Instance optimization (Sec. 2.2) also needs organ and lesion labels of the scans being registered; in the main experiments, IO uses only predicted labels, while the provided labels of validation and test scans are used for evaluation (Sec. 3.1). Organ labels are predicted from the CT by TotalSegmentator.[14] Lesion masks are predicted from PET/CT by an nnU-Net[16] set up as the winning autoPET III solution:[17] a residual-encoder U-Net (ResEnc-L) with 192^3 patches, initialized from their pretrained weights, with an auxiliary organ-segmentation head and misalignment augmentation. It was trained with patient-grouped five-fold cross-validation on all 494 scans of the non-test patients, including patients with a single examination. Validation scans are segmented by the fold model that did not see the patient, and test scans by the ensemble of the five fold models. For validation, this prevents registration and IO tuning from relying on lesion predictions made by a model trained on the same patient. The predicted lesion masks reach a mean Dice of 0.56 on the validation scans and 0.51 on the test scans (medians 0.69 and 0.67).

Preprocessing. For the network and IO, CT was clipped to $[-1000, 1500]$ HU and PET to $[0, 20]$ SUV, and both were rescaled to $[0, 1]$. As in the ConvexAdam baseline of the challenge,[18] the patient bed was removed from the CT. All PET quantities in the evaluation are computed on the original SUV images.

PET measures. The PSMAReg challenge uses metabolic tumor volume (MTV) and total lesion glycolysis (TLG) to measure how registration changes the moving scan's lesion volume and uptake.[2] For a PET image P in SUV, binary lesion mask M, and voxel volume v, they are

$$\mathrm{MTV}(M) = v \sum_x M(x), \qquad \mathrm{TLG}(P, M) = v \sum_x P(x) M(x). \tag{1}$$

Both quantities are computed on the same moving scan before and after warping it to the fixed frame; as in the challenge scorer, both PET and the lesion mask are warped by nearest-neighbor in-

terpolation. The reported preservation error for $Q \in \{\mathrm{MTV}, \mathrm{TLG}\}$ is its absolute relative change,

$$|\Delta Q| = \frac{|Q_{\mathrm{after}} - Q_{\mathrm{before}}|}{\max(Q_{\mathrm{before}}, 10^{-6})}. \tag{2}$$

This measures change induced by registration, not change in disease burden between examinations.

2.2 Registration Pipeline

The pipeline (Fig. 1) has three stages, all estimated from the moving scan (C_m, P_m) and the fixed scan (C_f, P_f), where C denotes CT and P PET. We denote the affine sampling transform by φ_{aff}, the network sampling transform by φ_θ, and their composition by φ. For each location x on the fixed grid, $\varphi(x)$ gives the location sampled in the original moving scan: the warped image is $\widetilde{C}_m(x) = C_m(\varphi(x))$, and the same convention applies to PET and labels.

Affine pre-registration. ANTs[19] estimates an affine transform from the two CTs, using body masks, a $[-1000, 1000]$ HU window, half-resolution images, and Mattes mutual information. It is converted to a dense displacement field.

Deformable network. We use LapIRN[12] with three pyramid levels at quarter, half, and full resolution. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level and integrated by scaling and squaring[20] into φ_θ. Each level receives four channels: the fixed CT and PET and the affinely aligned moving CT and PET. The levels are trained one after the other; a finer level is initialized from the coarser one, which is first kept frozen and then trained jointly. Each sub-network has 24 base channels and a trunk of ten residual blocks with an expansion factor of two.

Instance optimization. At test time, the deformation of each pair is refined by optimizing a residual stationary velocity field, initialized at zero and integrated by scaling and squaring (seven steps), on top of the total transformation φ. The residual is optimized at full resolution with Adam[21] (learning rate 0.025, 40 iterations). The objective is the same as in training (Sec. 2.3), with the predicted labels in place of ground-truth labels. Of the 40 iterates, we return the one with the lowest objective, in which the interpolated (soft) MTV and TLG terms are replaced by their nearest-neighbor (hard) values, i.e. by how the evaluation counts them; this selection uses only predicted labels.

2.3 Registration Objective

Network training and IO minimize a weighted sum of four groups of terms (Fig. 1):

$$\begin{aligned}\mathcal{L} = {} & \underbrace{\lambda_{\mathrm{NCC}}\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\mathcal{L}_{\mathrm{DSC}}}_{\text{registration accuracy}} + \underbrace{\lambda_{\mathrm{MTV}}\mathcal{L}_{\mathrm{MTV}} + \lambda_{\overline{J}}\mathcal{L}_{\overline{J}} + \lambda_J\mathcal{L}_J + \lambda_{\mathrm{TLG}}\mathcal{L}_{\mathrm{TLG}}}_{\text{PET quantification}} \\ & + \underbrace{\lambda_{\mathrm{rig}}\mathcal{L}_{\mathrm{rig}}}_{\text{bone rigidity}} + \underbrace{\lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\mathcal{L}_{\mathrm{NDV}}}_{\text{deformation regularization}}.\end{aligned} \tag{3}$$

All terms are evaluated on the total transformation φ. The PET-preservation terms and the rigidity term are the subject of this study; they can be switched on in network training, in IO, or in both (Sec. 3). The two coarse network levels never use them, because lesions and thin bones span only a few voxels at these resolutions.

Registration accuracy. $\mathcal{L}_{\mathrm{NCC}}$ is the negative local normalized cross-correlation of the CTs, summed over one to three scales depending on the pyramid level. PET is not used for similarity, because uptake changes between examinations with disease and therapy and a PET similarity would reward matching these changes. $\mathcal{L}_{\mathrm{DSC}}$ is a soft Dice loss on the CT organ labels.

PET quantification. The PET-preservation terms keep the PET measures of Eq. (1) unchanged by the registration. For the differentiable loss terms, let M be the moving lesion mask and $\widetilde{M}$, $\widetilde{P}$ the linearly warped mask and PET. Nearest-neighbor warping, as used in the evaluation, is piecewise constant in the displacement and gives no useful gradient; linear warping gives a soft lesion boundary instead. The volume and uptake terms penalize the relative change defined above, using the soft mask,

$$\mathcal{L}_{\mathrm{MTV}} = \left(\frac{\left| \sum_x \widetilde{M}(x) - \sum_x M(x) \right|}{\sum_x M(x) + \epsilon} \right)^2, \qquad \mathcal{L}_{\mathrm{TLG}} = \frac{\left| \sum_x \widetilde{P}(x)\widetilde{M}(x) - \sum_x P_m(x)M(x) \right|}{\sum_x P_m(x)M(x) + \epsilon}, \tag{4}$$

with $\epsilon = 10^{-5}$. Two further terms act on $\det J_\varphi(x)$, the local volume scale of the total affine-plus-deformable sampling transform, computed by central differences on the fixed voxel grid. A determinant of one preserves local volume; values above and below one correspond to expansion and contraction, respectively, of the sampled region in the moving scan (the opposite change in the displayed warped lesion). Let W_s and W_h be the linearly and nearest-neighbor warped moving lesion masks, respectively, on that grid. Then

$$\mathcal{L}_{\overline{J}} = \left(\frac{\sum_x W_s(x) \det J_\varphi(x)}{\sum_x W_s(x) + \epsilon} - 1 \right)^2, \qquad \mathcal{L}_J = \frac{\sum_x W_h(x) \big(\det J_\varphi(x) - 1 \big)^2}{\sum_x W_h(x) + \epsilon}. \tag{5}$$

$\mathcal{L}_{\overline{J}}$ constrains the mean volume scale over all lesions, but allows expansion in one region to cancel contraction in another; $\mathcal{L}_J$ penalizes these local changes separately. The mask W_s is held fixed when differentiating $\mathcal{L}_{\overline{J}}$, and the hard mask W_h is not differentiable with respect to displacement; both losses differentiate through the determinant.

Bone rigidity. Many PSMA lesions lie in bone, which moves rigidly between examinations. For each of the 61 skeletal labels of the fixed scan (in IO: the predicted labels), we fit a rigid transformation to the displacement of the structure in closed form[22,23] and penalize the mean squared residual of its voxels; $\mathcal{L}_{\mathrm{rig}}$ is the mean over structures with at least 50 voxels. Each bone is fitted independently, so neighboring bones may move relative to each other, and the term involves no finite differences, so it does not constrain the tissue around a bone.

Table 1 Loss weights of Eq. (3) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimization (IO).

	λ_{NCC}	λ_{DSC}	λ_{MTV}	$\lambda_{\overline{J}}$	λ_{J}	λ_{TLG}	λ_{rig}	λ_{smooth}	λ_{NDV}
Training	5	5	5	0.125	1.25	1.25	0.05	10	10^4
IO	5	5	$4 \cdot 10^4$	10^3	4	16	0.25	0	10

Deformation regularization. $\mathcal{L}_{\text{smooth}}$ is the mean squared spatial gradient of the displacement. $\mathcal{L}_{\text{NDV}}$ is a differentiable version of the challenge's non-diffeomorphic volume, which splits every voxel into tetrahedra and accumulates the negative part of their Jacobian determinants.[24]

Weights. Table 1 lists the weights. In training, the alignment weights are the same for all models, and the PET and rigidity weights are either used or set to zero. The IO weights were tuned on the validation set (Sec. 3.3).

2.4 Rigidity Penalties Compared

Besides the per-bone rigid fit, we evaluate the rigidity penalty of Staring et al.,[25] which requires the Jacobian J inside a bone to be a rotation: $\det J = 1$, $J^\top J = I$, and vanishing second derivatives of the deformation. It is computed with finite differences, whose stencil reads the 6 face and 12 edge neighbors of a voxel. At a bone boundary, part of the stencil lies outside the bone, so the penalty also acts on the surrounding soft tissue or on a neighboring bone. A common remedy is to erode the bone mask so that the stencil stays inside; we erode each bone by exactly this 18-neighbor footprint, which removes the leakage completely. The three variants (Staring, Staring with erosion, per-bone rigid fit) are compared in IO (Sec. 3).

2.5 Training

Each level was trained with Adam[21] for 10^5, 10^5, and $1.2 \cdot 10^5$ steps (quarter, half, full resolution) with learning rates of $1.5 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, and 10^{-5}, a batch size of one with gradients accumulated over four pairs, a linear warm-up over five epochs, and a cosine decay to 10 % of the initial rate. A finer level is trained jointly with the coarser ones after ten epochs. Convolutions run in bfloat16, all losses in single precision. Training took 18 h, 22 h, and 3.7 days for the three levels on one NVIDIA A100 (80 GB). The augmentations mimic longitudinal variability: left–right flipping ($p = 0.5$), CT intensity shifts of ± 0.02 in the normalized window (about ± 50 HU), CT scaling between 0.9 and 1.1, PET scaling between 0.85 and 1.15, and field-of-view mismatch by cropping up to 40 axial slices from both scans and up to 10 further slices from each scan separately. The two coarse levels are shared by all models; the variants of this study differ only in the losses of the full-resolution level. Checkpoints were selected on the validation set: the coarse levels by CT Dice, the full-resolution level by the best accuracy (CT Dice and HD95) among checkpoints whose validation folding stays below $1.4 \cdot 10^{-4}$ % non-diffeomorphic volume. We did not select on MTV or TLG, so that the PET losses are not credited with effects of the checkpoint choice.

Use of AI tools. Parts of the code for training, evaluation, statistical analysis, and figure generation were written with the help of an AI coding assistant. All generated code was reviewed and tested by the authors, who are responsible for the analyses and results.

3 Experiments

3.1 Evaluation Metrics

All metrics are computed with the provided labels of the test scans and follow the scorer of the challenge: the moving scan is warped with the voxel displacement field, sampled with corner-aligned grid coordinates, and PET and labels are warped by nearest-neighbor interpolation. The challenge submits displacement fields at half resolution and upsamples them; we evaluate the full-resolution fields directly. Of the 28 test patients (38 pairs), 26 patients (30 pairs) have lesions in the moving scan and contribute to the PET metrics.

Alignment. CT Dice and the 95th percentile of the Hausdorff distance (HD95), averaged over the 117 organ labels.

PET-preservation metrics. We compute $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ as defined in Eq. (2), using the provided moving-scan lesion masks and the challenge scorer. "Initial" denotes the unregistered scans; their MTV and TLG change is zero by construction.

Per-lesion change. The aggregate errors of Eq. (2) can hide large changes of individual lesions that cancel. We therefore also split the moving lesion mask into 26-connected components and compute $|\Delta\mathrm{MTV}|$ for each. The 30 pairs with lesions contain 1435 components. For the per-lesion size and location analysis, we exclude 108 components smaller than five voxels: a one-voxel difference changes their volume by at least 25 %, making their relative changes highly quantized. This leaves 1327 lesions, grouped by size (small < 0.5 ml, medium < 5 ml, large) and location (bone-associated if at least half of the lesion lies on a skeletal label). Scan-level PET metrics and the lesion-coverage analysis retain all 1435 components so that the coverage decomposition includes the full evaluated lesion volume.

Deformation quality. Percentage of non-diffeomorphic volume (NDV) as defined by the challenge,[24] within the challenge's mask (voxels where the fixed CT exceeds its minimum intensity).

Bone metrics. For the rigidity comparison, we additionally compute Dice and HD95 separately over bones and soft-tissue organs. We measure within-bone distance distortion as the mean relative change in physical distance between randomly sampled pairs of points in the same fixed-scan bone, including its boundary voxels. We compute this separately for short (5–15 mm) and long (≥ 15 mm) distances; pairs below 5 mm are excluded because their relative change is sensitive to interpolation and discretization at this image resolution. Short distances characterize local deformation, whereas long distances characterize deformation across a larger part of the bone. We also compute NDV inside bones and in a three-voxel band around them and, as a secondary measure, the residual of a rigid fit to each bone's displacement (bone rigidity error). We also measure the fraction of stencil reads that leave the bone of the center voxel (stencil leakage) and the fraction

of bone voxels and of bones that remain after erosion for the eroded Staring penalty. Erosion does not restrict the point pairs used for distance distortion.

3.2 Statistical Analysis

A patient may contribute several pairs and many lesions. We average each metric over a patient's eligible pairs before analysis: CT Dice, HD95, and NDV comparisons include 28 test patients, whereas MTV and TLG comparisons include the 26 patients with moving-scan lesions. Paired comparisons use only patients with values for both methods and a two-sided Wilcoxon signed-rank test;[26] the implementation doubles the smaller one-sided p-value, drops zero paired differences, assigns average ranks to tied nonzero magnitudes, and returns $p = 1$ when all differences are zero. Holm correction[27] is applied separately per metric to three comparisons against "None" (no losses) in each ablation block, three IO-only versus training-plus-IO comparisons, four leave-one-out comparisons, and comparisons of our method against the other baseline-table methods (nine for CT, HD95, and NDV; eight for MTV and TLG, because "Initial" has zero PET change by construction). We call a difference significant if its adjusted $p < 0.05$. Confidence intervals are 95 % percentile patient-bootstrap intervals:[28] scan-level plots show intervals for method means, whereas intervals stated for IO-only minus training-plus-IO are for paired patient-mean differences. For per-lesion medians, patients are resampled together with all their lesions (cluster bootstrap).

For the comparison with baselines, we additionally rank the nine resampled methods with the significance-based ranking of the challenge: per metric, all 72 directed method pairs are compared with one-sided Wilcoxon tests and Holm correction, and a method scores (number of methods it significantly outperforms $+1$)/K for K methods. Scores are averaged into accuracy (CT Dice, HD95), PET preservation ($|\Delta\mathrm{MTV}|$, $|\Delta\mathrm{TLG}|$), and regularity (NDV, after subtracting a practical-equivalence threshold of 0.005 % per pair), and combined into $100 \cdot \mathrm{acc}^{0.4}\,\mathrm{PET}^{0.4}\,\mathrm{reg}^{0.2}$. The challenge additionally scores Dice and HD95 of organs with physiological PET uptake, for which no labels are available to us; we omit them. Dice or HD95 of lesions would not be meaningful either, because lesions appear, disappear, and change between examinations. The stability of the ranks is shown by 95 % rank intervals over 1000 patient bootstrap samples.

3.3 Tuning and Model Selection

All tuning used the validation set. For our IO settings and the choice of our method, we use one selection rule: among the candidates of a grid whose CT Dice is within two points of the most accurate one, we keep the one with the lowest $|\Delta\mathrm{MTV}| + |\Delta\mathrm{TLG}|$. We do not use the challenge ranking for this, because it is decided by accuracy: differences in Dice between candidates are consistently significant, differences in MTV and TLG rarely, so the ranking almost always keeps the most accurate candidate regardless of preservation. The baselines were tuned with the challenge ranking, ranking the candidates of each method's grid against each other.

- **IO:** on the network trained with both losses, step-wise: first learning rate $\times$ iterations ($\{0.0125, 0.025, 0.05, 0.1\} \times \{10, 20, 40\}$), then a common scale of the four PET weights ($\times\{0, 0.1, 0.25, 0.5, 1, 2, 4\}$ of the training-derived values); chosen: learning rate 0.025 with 40 iterations, and $\times 0.1$ (Table 5). Neither the selection rule nor the ranking rewards rigidity, which costs Dice without reducing the PET metrics or folding, so both would set λ_{rig} to zero; we fixed $\lambda_{\mathrm{rig}} = 0.25$ instead.

- **NiftyReg**:[29,30] `reg_aladin` followed by `reg_f3d` on the GPU with normalized mutual information (local NCC is not available on the GPU), over the bending-energy weights $\{0.0002, 0.0005, 0.001, 0.005, 0.02\}$ and final control-point spacings of $\{3, 5, 7, 9\}$ voxels; chosen: 0.0002 and 9 voxels.

- **ConvexAdam**:[18] after its own ANTs affine registration with a $[-300, 300]$ HU CT window, over the regularization weight $\lambda \in \{0.1, 0.25, 0.5, 1.25, 2, 4\}$; chosen: 0.1.

Several choices ended at the edge of their grid; extending the grids changed the validation scores only marginally.

Choice of our method. Before the experiments, we defined our full method as the network trained with both losses followed by IO with both losses. Instead, we select our method on the validation set among the IO variants of Sec. 3.4 and IO directly after the affine pre-registration, with the same rule as for the IO settings: the variant with the lowest $|\Delta\mathrm{MTV}| + |\Delta\mathrm{TLG}|$ among those within two Dice points of the most accurate one. This selected the network trained without the losses followed by IO with both losses (79.2 % Dice, $|\Delta\mathrm{MTV}|$ 6.9 %, $|\Delta\mathrm{TLG}|$ 8.9 %; the most accurate variant, IO without the losses, reached 80.8 %). The variants with the losses in training and IO changed the lesions slightly less (e.g. 5.8 % and 7.9 % with both losses) but were 2.6 points less accurate than the most accurate variant. We refer to the selected variant as our method and report the predefined full method and IO without the losses alongside it in the comparison with baselines.

3.4 Comparisons

Effect of the losses. Four networks were trained with the same alignment and regularization terms and with neither, the PET-preservation, the rigidity, or both groups of losses. Each is evaluated as network output and with IO using the same losses as in its training.

Losses in training or only in IO. The network trained without the losses is combined with IO using PET preservation, rigidity, or both, and compared with the networks trained with these losses (and IO with the same losses). To measure what the network contributes, IO with both losses is also run directly after the affine pre-registration.

Contribution of each PET term. On the network trained without the losses, IO with all four PET terms is compared with IO in which one term at a time is switched off.

Per-lesion terms. Because the PET losses constrain the total volume and uptake of a scan, we test per-lesion versions of the volume, uptake, and mean-Jacobian terms in IO, added to the PET terms on the network trained without the losses. The predicted lesion mask is split into connected components; each of the eight largest components with at least 20 voxels gets its own relative volume and uptake change and mean Jacobian determinant, and the terms are averaged over the components weighted by size, so that changes of different lesions cannot cancel. Their weight is selected on the validation set among $\{1, 10, 100\}$ times a base value scaled like the other PET weights, with the selection rule of Sec. 3.3 applied to the median per-lesion $|\Delta\mathrm{MTV}| + |\Delta\mathrm{TLG}|$, the quantity these terms target.

Comparison with baselines. Initial alignment, the affine pre-registration, NiftyReg, ConvexAdam, IO after the affine, and our variants are compared on the test set. All registrations use the same follow-up-to-baseline pairs and are scored with the same provided organ and lesion labels and original-SUV PET images. NiftyReg and ConvexAdam use CT alone for registration, with no organ or lesion labels; both remove the patient bed, but each uses its own affine step and CT preprocessing (NiftyReg: `reg_aladin` on bed-removed CT; ConvexAdam: ANTs with a $[-300, 300]$ HU window). Our method uses CT and PET, the affine pre-registration described above, provided labels in network training, and predicted labels during IO. The NiftyReg and ConvexAdam settings were selected on the validation pairs by the challenge ranking, with candidates ranked within each method's grid (Sec. 3.3). Runtimes are per pair on one NVIDIA RTX A6000 GPU and include the affine step.

Rigidity penalties. On the network trained with the PET losses, IO with the PET terms is combined with no rigidity, Staring, Staring with erosion, or the per-bone rigid fit. Because the penalties have different units, a reference weight is first calibrated for each so that its initial value on five validation pairs equals that of the rigid fit at $\lambda_{\text{rig}} = 0.25$ (Staring 0.66, Staring with erosion 1.03). Each method is then evaluated on the validation set at $\{0.1, 0.3, 1, 3, 10, 30, 100\}$ times its reference weight, giving a curve of CT Dice against within-bone distance distortion. For the test set, we choose for each method the largest weight whose validation Dice is within two Dice points of the variant without rigidity, the same tolerance as in the selection rule above, with the grid capped at $\times 100$: $\times 3$ for Staring, and $\times 100$ for Staring with erosion and the rigid fit.

Accuracy–preservation trade-off. On the network trained without the losses, IO with the PET terms (without rigidity) is run with all four PET weights scaled by $\{0, 10^{-4}, 3{\cdot}10^{-4}, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3$ relative to the training-derived values; $\times 0.1$ is the setting chosen by the IO tuning. Besides the evaluation on the ground-truth labels, we compute CT Dice, $|\Delta\text{MTV}|$, and $|\Delta\text{TLG}|$ on the predicted labels that IO optimizes, with the same definitions. To probe the effect of lesion-mask quality, we additionally rerun IO with the provided lesion masks instead of the predicted ones (organ labels remain predicted), at the tuned and full PET weights. This diagnostic intervention uses labels unavailable in the intended test-time setting.

Lesion coverage in predicted masks. Each ground-truth lesion of the moving scan is assigned to a group by the fraction of its voxels covered by the predicted lesion mask: missed (0 %), partly covered ($< 50\,\%$), or covered ($\geq 50\,\%$). For each group, we report two quantities, both computed per scan as in Eq. (2) and averaged over patients: the change within the group, using only the group's lesions, and the contribution of the group to the evaluation metric, i.e. the change if only the group's lesions changed and all other lesions kept their volume and uptake. Since changes of different lesions in a scan partly cancel, the contributions add up to slightly more than the evaluation metric.

4 Results

All results are on the test set (28 patients, 38 pairs); values are mean±SD over patients, and differences are given in percentage points. Except where stated, IO uses predicted masks, while the reported alignment and PET-preservation metrics use the provided labels. We first ask whether the

losses alter the balance between alignment and scan-level PET change, and whether they should act during training or only during IO. We then test whether that scan-level result extends to individual lesions, compare the selected method with baselines, and examine the effects of PET-loss weights and predicted-mask coverage. The rigidity comparison separately tests how well the penalties preserve bone geometry.

4.1 Effect of the PET-Preservation and Rigidity Losses

Figure 2 compares four configurations (the full results with significance tests are in Table 4, Appendix A): no PET-preservation or bone-rigidity loss, PET preservation alone, rigidity alone, and both. Each network is evaluated before IO (squares; $\times$ without losses) and after IO (filled circles; $+$ without losses). In IO, the PET-preservation and bone-rigidity losses are enabled or disabled to match that network's training configuration. Without IO (squares vs. $\times$), training with the PET losses lowered the volume change of the lesions from 10.3 % to 8.2 % (significant) and the uptake change from 12.5 % to 9.8 % (not significant), at the cost of 1.9 Dice points. IO raised the Dice of every network by 3.7 to 4.7 points (each square to the filled circle of the same color, $\times$ to $+$). With the PET losses in training and IO (blue filled circle vs. $+$), $|\Delta\mathrm{MTV}|$ was 8.2 % instead of 11.3 % and $|\Delta\mathrm{TLG}|$ 10.3 % instead of 14.5 %, for 2.5 Dice points; with IO, these reductions were not significant. The rigidity loss alone (red) did not change the Dice of the network and cost 0.6 points with IO; it lowered $|\Delta\mathrm{MTV}|$ significantly for the network output (10.3 % to 8.9 %) and $|\Delta\mathrm{TLG}|$ significantly with IO (14.5 % to 10.6 %). Combining both groups (purple) preserved about as well as the PET losses alone and was the least accurate variant (73.6 % and 77.3 % Dice). HD95 followed the Dice in every case (network 7.2 to 7.4 mm, with IO 6.6 to 7.1 mm). All network outputs and the IO variants without the PET losses stayed below the challenge's NDV threshold of 0.005 % (50 ppm) on every pair; with the PET losses in IO, one pair each exceeded it (150 and 262 ppm), and the variant means were at most 10 ppm.

The losses therefore change the accuracy–preservation balance; the next question is whether they need to shape the trained network or can act only during per-pair refinement.

4.2 Losses in Training or Only in IO

To test whether the losses need to act during training, Fig. 2 compares, for each loss group, the network trained with the losses followed by IO with them (filled circles) with the network trained without them followed by the same IO (hollow circles); the second dashed segment of each color connects the two. With both loss groups (purple), applying them only in IO increased Dice by 1.0 points (95 % CI 0.4 to 1.6) compared with using them in training and IO. The paired differences (IO only minus training and IO) were $+0.8$ points in $|\Delta\mathrm{MTV}|$ (95 % CI -0.4 to 2.0) and $+1.0$ points in $|\Delta\mathrm{TLG}|$ (-0.3 to 2.6). With PET losses alone (blue), the Dice gain was 0.9 points (0.1 to 1.7), while $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ increased by 1.1 (-0.3 to 2.6) and 1.5 (-0.0 to 3.4) points. With rigidity alone (red), the paired differences were 0.0 (-0.0 to 0.1) Dice points, $+0.7$ (-0.0 to 1.8) MTV points, and $+0.8$ (0.0 to 1.9) TLG points. None of the preservation differences was significant after Holm correction; the intervals do not establish equivalent preservation. Without IO, training with the losses reduced registration-induced PET changes in the network-only variants tested (Fig. 2, squares; Table 4).

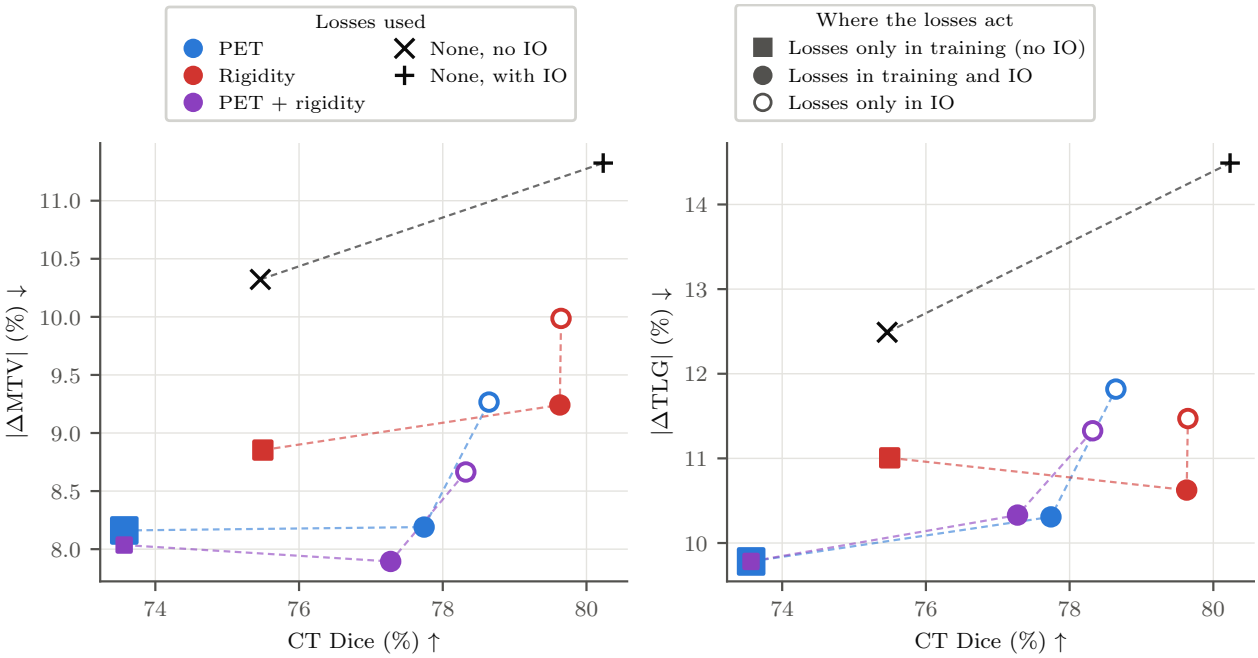


Fig 2 Effect of the PET-preservation and rigidity losses and of where they act (test set; CT Dice: 28 patients, PET: 26 patients). Color: losses used; marker: where they act (square: only in network training, network output without IO; filled circle: in training and IO; hollow circle: only in IO, on the network trained without losses). Without losses: network output (×) and with IO (+). Dashed lines connect the variants of a loss group. Points: patient means; the significance tests are reported in the text and in Table 4.

4.3 Per-Lesion Volume Change

To reveal changes hidden by scan-level aggregation, Fig. 3 shows the volume change of 1327 lesions from 26 patients, after excluding 108 components smaller than five voxels. Small lesions changed most in every condition: for bone-associated lesions with IO and without the losses (+), the median change was 30 % (small), 19 % (medium), and 10 % (large). A small lesion has fewer than about 20 voxels, so gaining or losing one or two voxels is already a large relative change. The per-lesion uptake change followed the volume change in all strata (Fig. 10, Appendix B). Without the losses, IO increased the distortion of lesions (× to +), most of all of small lesions (soft tissue from 17 % to 33 %, bone-associated from 25 % to 30 %); bone-associated lesions tended to grow (median signed change +1 to +5 %) and soft-tissue lesions to shrink (−3 to −14 %). The rigidity loss (red) acted mainly on bone-associated lesions, as expected from its construction: with IO, it lowered their median change for small, medium, and large lesions from 30 % to 20 %, 19 % to 12 %, and 10 % to 5 %, and left medium and large soft-tissue lesions unchanged (19 % and 17 %). The PET losses (blue) acted in both locations, already without IO (squares), and with IO, for example from 19 % to 10 % for medium soft-tissue lesions and from 30 % to 16 % for small bone-associated lesions. With both groups (purple), bone-associated lesions changed by 14 %, 9 %, and 6 %, similar to the PET losses alone; only for large bone-associated lesions, the rigidity loss alone changed them less (5 %). With the losses, IO did not reduce the per-lesion change below that of the network output (circles vs. squares of the same color), in line with the unchanged scan-level change (Sec. 4.1). The PET losses in IO penalize the change in the total volume and uptake of the predicted lesion mask, so the growth of some lesions can offset the shrinkage of others. Even with

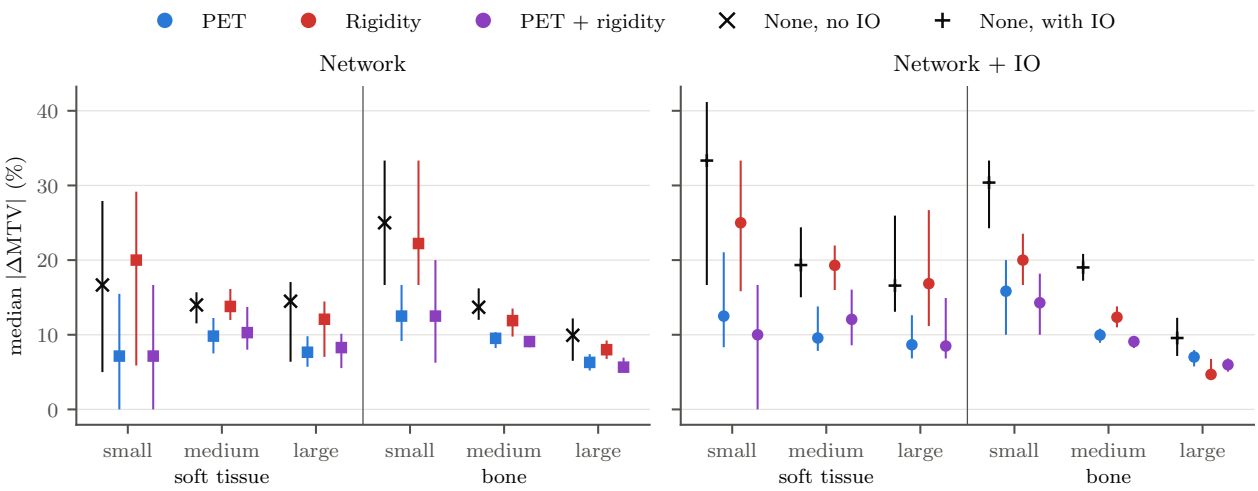


Fig 3 Per-lesion volume change on the test set (1327 lesions of 26 patients after excluding 108 components smaller than five voxels), by location (soft tissue / bone-associated, ≥50 % of the lesion on bone) and size (small <0.5 ml, medium <5 ml, large). Median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: with instance optimization (IO) using the same losses (squares / ×: without IO; circles / +: with IO; colors as in Fig. 2). Error bars: 95 % confidence intervals from a patient-cluster bootstrap.

the ground-truth lesion masks in IO, which lowered the scan-level $|\Delta\mathrm{MTV}|$ to 0.1 % (Sec. 4.7), the median per-lesion change was 13.5 %, as with the predicted masks (13.7 %).

Adding per-lesion terms to IO did not change this either. On the validation set, the selection chose ten times the base weight (median per-lesion $|\Delta\mathrm{MTV}|$ 11.3 % vs. 12.5 % without the terms). On the test set, the median per-lesion volume change fell only from 13.7 % to 13.3 %, it decreased for only 5 of 26 patients, and small lesions changed more (median 20.0 % vs. 16.7 %); the scan-level changes were unchanged ($|\Delta\mathrm{MTV}|$ 9.3 %, $|\Delta\mathrm{TLG}|$ 12.0 % vs. 11.8 %), and Dice rose by 0.6 points. At the base weight, the terms also had no effect with the ground-truth lesion masks (median 13.5 % with and without them). The medians of small lesions take only a few distinct values (e.g. 12.5 %, 14.3 %, 33.3 %) because they are ratios of small voxel counts.

4.4 Contribution of Each PET Term

To identify which constraints account for preservation in IO, we remove one PET term at a time from IO on the network trained without the losses (Table 2). The mean-Jacobian term $\mathcal{L}_{\overline{J}}$ carried the preservation: without it, $|\Delta\mathrm{MTV}|$ increased from 9.3 % to 10.4 % and $|\Delta\mathrm{TLG}|$ from 11.8 % to 13.5 % (both significant). Without the voxel-wise term $\mathcal{L}_J$, $|\Delta\mathrm{TLG}|$ increased to 12.3 % (significant), whereas $|\Delta\mathrm{MTV}|$ was unchanged (9.3 %). The volume and uptake terms worked against preservation. Without $\mathcal{L}_{\mathrm{MTV}}$, Dice increased from 78.6 % to 79.3 % and $|\Delta\mathrm{TLG}|$ decreased to 10.7 % (both significant; $|\Delta\mathrm{MTV}|$ 8.6 %, not significant). Without $\mathcal{L}_{\mathrm{TLG}}$, $|\Delta\mathrm{TLG}|$ decreased to 9.8 % (significant), while Dice (78.7 %) and $|\Delta\mathrm{MTV}|$ (8.6 %) did not change significantly. Both terms measure the change on the linearly warped, soft mask and PET (Eq. 4), whereas the evaluation warps both by nearest neighbor; lowering the soft change therefore need not lower the evaluated one.

Table 2 Contribution of each PET term in IO (test set; CT Dice: 28 patients, PET: 26 patients): network trained without PET losses (T1), IO with all PET terms or with one term removed. Mean±SD over patients. †/‡: significantly better/worse than with all terms (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over four comparisons within each column, $p < 0.05$).

IO PET terms	Dice (%) ↑	$\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓	$\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓
All PET terms	78.6±4.4	9.3±11.6	11.8±11.7
without $\mathcal{L}_{\mathrm{MTV}}$	79.3±4.6†	8.6±11.6	10.7±11.9†
without $\mathcal{L}_{\overline{J}}$	78.7±4.5	10.4±11.1‡	13.5±11.2‡
without $\mathcal{L}_{\mathrm{TLG}}$	78.7±4.4	8.6±11.7	9.8±12.5†
without $\mathcal{L}_{J}$	78.6±4.5	9.3±11.4	12.3±11.5‡

4.5 Comparison with Baselines

To place our method in context, we compare it with CT-only registration baselines, with IO applied directly after affine pre-registration, and with our other variants (Table 3). Our method, with the losses only in IO, reached 78.3 % CT Dice and 6.89 mm HD95, significantly better than NiftyReg (70.1 %, 8.85 mm), ConvexAdam (61.7 %, 9.16 mm), and the affine pre-registration (46.4 %, 14.00 mm); its $\lvert\Delta\mathrm{MTV}\rvert$ (8.7 %) and $\lvert\Delta\mathrm{TLG}\rvert$ (11.3 %) did not differ significantly from those of NiftyReg (8.7 %, 10.2 %) and ConvexAdam (9.3 %, 10.8 %). Compared with the same network before IO, our method gained 2.9 Dice points (95 % CI 2.3 to 3.4). The paired changes in $\lvert\Delta\mathrm{MTV}\rvert$ and $\lvert\Delta\mathrm{TLG}\rvert$ were -1.7 points (95 % CI -3.6 to 0.3) and -1.2 points (-3.6 to 1.2), respectively, so an increase in scan-level PET change was not detected. NiftyReg, tuned on the validation set, exceeded the folding threshold on 5 of 38 validation pairs and 8 of 38 test pairs (21 %), with up to 0.37 % NDV; ConvexAdam stayed below it on every pair. Our method exceeded it on two test pairs (76 and 125 ppm), and the variant with the losses also in training on one (262 ppm). IO directly after the affine pre-registration reached only 56.6 % Dice: within the same iteration budget, IO cannot replace the network. Resampling alone already changes the measurements: the affine pre-registration changed MTV by 4.5 % and TLG by 4.4 %, the lowest of all methods.

Among our variants, IO without the losses was the most accurate (80.2 % Dice, significantly higher than our method) but changed the lesions most ($\lvert\Delta\mathrm{MTV}\rvert$ 11.3 %, $\lvert\Delta\mathrm{TLG}\rvert$ 14.5 %). The losses only in IO lowered these changes to 8.7 % and 11.3 % at a cost of 1.9 Dice points, and the losses also in training to 7.9 % and 10.3 % at a cost of 2.9 points; after correction for multiple comparisons, none of these differences in preservation was significant. By the challenge's composite score, IO without the losses ranked first, our method second, and the variant with the losses in training and IO third, with overlapping bootstrap rank intervals (1–4, 1–4, and 1–5). The ranking was decided by accuracy: apart from the affine pre-registration on $\lvert\Delta\mathrm{TLG}\rvert$, no method won a significant comparison in preservation or folding. The network alone needs 11 s per pair, as fast as ConvexAdam and ten times faster than NiftyReg (108 s); IO adds about 2.5 minutes. Figure 4 shows a test pair whose lesion volume grew tenfold between baseline and follow-up (from 2400 to 24 191 voxels); it was chosen for this change, not for its registration result, and illustrates alignment. The scans were initially far apart (CT Dice 3.5 %); our method aligned them to 83.7 % Dice (NiftyReg 70.8 %, ConvexAdam 65.7 %), and NDV remained below the challenge threshold. The volume and uptake of the 163 follow-up lesions changed by $+2.6\,\%$ and $+6.6\,\%$, more than with IO without the losses ($-0.6\,\%$ and $+2.3\,\%$). Before registration, the CT overlay (Fig. 4c) and the label residual (e) show misalignment across the whole trunk; after registration (d, f), the residual is reduced to thin bands at organ boundaries and to parts of the lumbar vertebrae, and the warped le-

Table 3 Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs; PET: 26 patients, 30 pairs). Values are mean±SD over patients (a patient's eligible pairs are averaged first). Rank: descriptive challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimization (IO); IO "–": no IO, "basic": IO with the similarity and regularization terms only. †/‡: Ours + IO with PET + rig. only in IO (our method, chosen on the validation set) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over nine comparisons for CT/HD95/NDV and eight for PET within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). $>$ thr.: pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000 (measured in the runs with the earlier scorer, same IO settings), including the affine step. Affine + IO: our instance optimization directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.

Method	Training	IO	Rank ↓	Dice (%) ↑	HD95 (mm) ↓	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓	NDV (ppm) ↓	> thr. (%)	Runtime (s)
Initial	–	–	–	9.5±8.1†	44.37±21.32†	0.0±0.0	0.0±0.0	0±0	0	–
Affine	–	–	8 [4–9]	46.4±11.0†	14.00±4.83†	**4.5**±4.6	**4.4**±4.6‡	**0**±0	0	10.4±0.3
NiftyReg	–	–	6 [1–8]	70.1±5.0†	8.85±3.73†	8.7±12.8	10.2±16.9	328±885	21	107.7±24.9
ConvexAdam	–	–	7 [7–9]	61.7±4.5†	9.16±3.14†	9.3±12.0	10.8±16.5	**0**±0	0	11.3±0.6
Affine + IO	–	PET + rig.	8 [4–9]	56.6±12.1†	12.88±4.81†	5.3±4.1	6.8±4.0	0.45±0.31	0	155.5±2.6
Ours	–	–	4 [3–6]	75.5±4.7†	7.20±2.30†	10.3±9.5	12.5±11.1	0.056±0.057	0	11.0±0.4
Ours	PET + rig.	–	5 [1–6]	73.6±4.6†	7.44±2.37†	8.0±9.0	9.8±10.9	0.64±0.62	0	11.3±0.3
Ours + IO	–	basic	1 [1–4]	**80.2**±4.2‡	**6.64**±2.28‡	11.3±11.9	14.5±14.1	0.65±0.13	0	153.9±2.6
Ours + IO	PET + rig.	PET + rig.	3 [1–5]	77.3±4.9†	7.07±2.38†	7.9±9.7	10.3±10.4	10.4±49.3	3	156.6±2.7
Ours + IO	–	PET + rig.	2 [1–4]	78.3±4.4	6.89±2.31	8.7±10.7	11.3±11.2	4.15±9.45	5	157.2±2.6

sion contours lie in the corresponding bones (h). The Jacobian determinant (g) shows fine-grained changes in the local volume scale of the sampling transform throughout the body, which IO adds to the smooth network field because it has no smoothness term ($\lambda_{\text{smooth}} = 0$).

4.6 Rigidity Penalties

Bones should retain their internal geometry during registration even when they move relative to one another. Because label overlap does not measure this property, we compare rigidity penalties by how well they preserve within-bone distances and report the accompanying changes in bone and soft-tissue alignment. In the finite-difference Staring penalty, 21.2 % of the stencil reads leave the bone of the center voxel (all 560 labeled challenge scans); eroding the bones removes this leakage but keeps only 35.5 % of the bone voxels and removes 23 % of the bones entirely (ribs: 2.4 % of the voxels kept, 58 % of the ribs removed).

On the validation set (Fig. 5), the per-bone rigid fit reached a lower within-bone distortion than the Staring penalty at every Dice level, most clearly for long distances, where Staring did not get below 2.8 % while the rigid fit reached 1.8 %. With erosion, the Staring penalty barely changed the distortion even at $\times 100$ (short distances 8.7 % vs. 12.7 % without rigidity), and Dice stayed almost constant, because most of the skeleton is no longer constrained. Within the two-point tolerance, the selection chose $\times 3$ for Staring and the largest grid value, $\times 100$, for the other two penalties.

On the test set, Fig. 6a shows each rigidity measure as a percentage of its value with IO without rigidity (dashed line at 100 %). Each penalty was best on the measure it optimizes: the rigid fit lowered the rigid-fit residual to 53 %, Staring the Staring energy to 18 %. On the distance distortion, which neither penalty optimizes, the rigid fit was best, with 41 % (short) and 42 % (long) of the distortion without rigidity, compared with 47 % and 73 % for Staring and 69 % and 80 % for

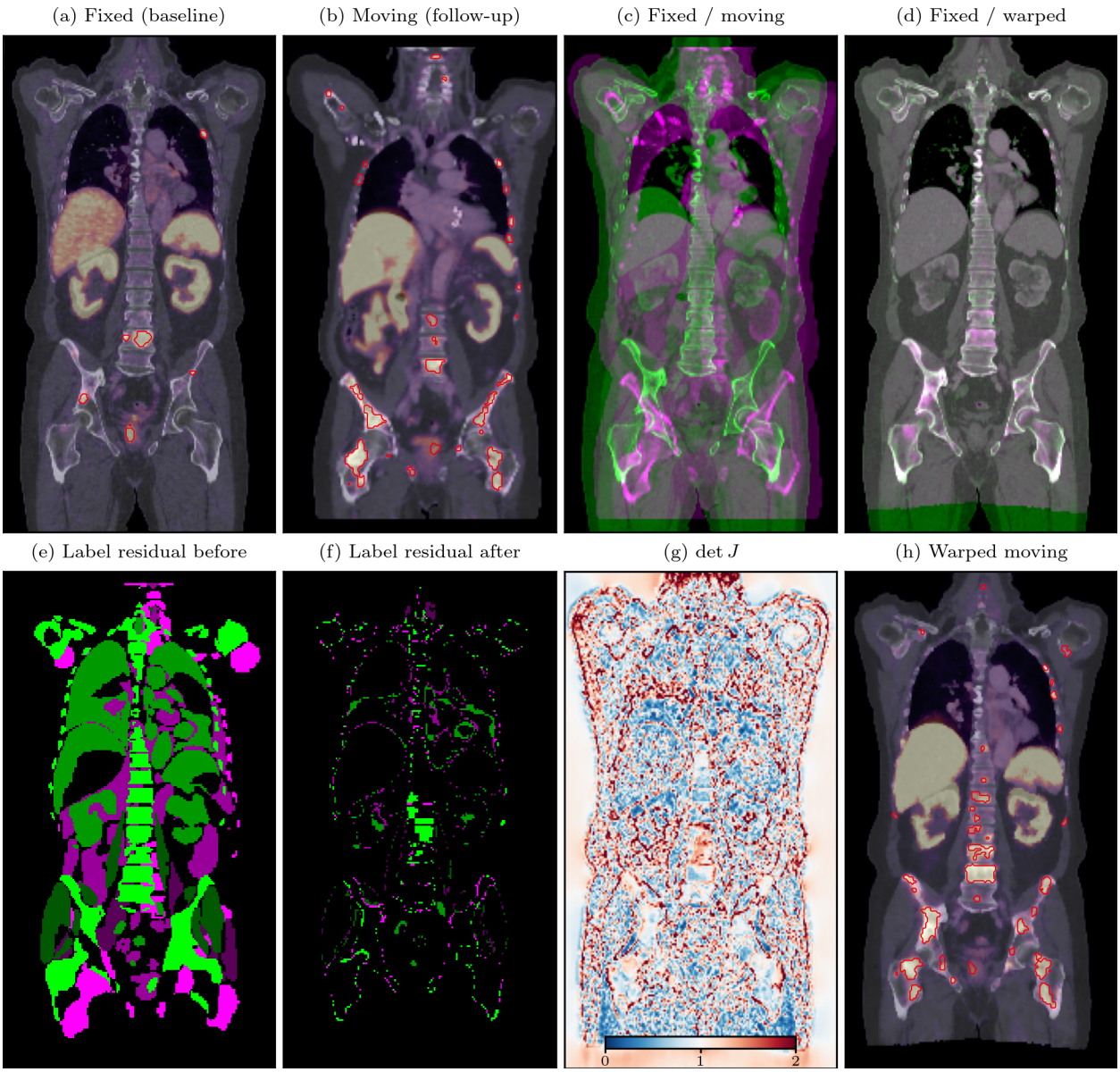


Fig 4 Test pair with a large change in tumor volume (patient 0146; lesion volume 2400 voxels at baseline, 24191 at follow-up), registered by our method (PET and rigidity losses only in IO); one coronal slice. (a) Fixed and (b) moving CT with PET (SUV 0–8, 35 % opacity) and ground-truth lesion contours (red). (c, d) CT of the fixed scan (green) overlaid with the moving scan (magenta) before (c) and after (d) registration; gray where they agree. (e, f) Organ labels that differ before (e) and after (f) registration: green where the fixed scan has a label that the moving scan lacks there, magenta where only the moving scan is labeled; bright, medium, and dark shades for bones, organs, and muscles. (g) Jacobian determinant of the fixed-to-moving sampling transform: white is volume-preserving (1), blue is below 1 and red is above 1; changes in the displayed warped anatomy have the opposite direction. (h) Warped moving CT with PET and warped lesion contours.

Staring with erosion. Fig. 6b shows the cost in alignment: the rigid fit and Staring lowered CT Dice by 1.6 and 2.0 points and bone Dice by about 4 points, whereas soft-tissue Dice was unchanged with Staring and rose by 0.9 points with the rigid fit; Staring with erosion cost almost no Dice but also constrained the bones least.

In absolute terms (Table 9, Appendix F), the rigid fit reduced the short-distance distortion from 13.4 % to 5.0 % and the long-distance distortion from 4.4 % to 1.8 %; all three penalties improved every rigidity measure significantly over IO without rigidity, and all pairwise differences in distortion between the penalties were significant. Staring also removed folding inside bones (NDV in bone from 6.5 to 0.3 ppm), whereas the rigid fit, which constrains only the displacement, left it unchanged (9.6 ppm, not significant).

When added on top of the PET losses, none of the three rigidity penalties produced a detected additional change in lesion volume or uptake preservation on the test set (Table 9). This comparison differs from Fig. 2, where rigidity used without the PET losses significantly reduced network-output $|\Delta\mathrm{MTV}|$ and IO $|\Delta\mathrm{TLG}|$ relative to the corresponding unconstrained variants. On the validation set (Fig. 7), every penalty and weight changed the lesions less than IO without rigidity, but the change did not decrease with the weight; the drop from weight 0 to the smallest weight came mostly from two patients whose volume change fell by 27 and 10 points already with the rigid fit at $\times 0.1$; without these two patients, all weights stayed within about one point of IO without rigidity ($|\Delta\mathrm{MTV}|$ 3.7 to 4.9 % vs. 4.4 %). Per lesion, the penalties lowered the median volume change of bone-associated lesions only from 9.3 % to 7.7–8.8 %, whereas an exactly affine transformation (the affine pre-registration) changed them by a median of 1.0 %.

4.7 Accuracy–Preservation Trade-off

We vary the PET-loss weights in IO to show how alignment and lesion preservation change together and whether stronger weights resolve the residual error. Figure 8 varies the PET weights in IO on the network trained without the losses. Already the smallest factor ($\times 10^{-4}$) lowered the volume change of the predicted lesion masks from 9.3 % (IO without the PET losses) to 2.1 % and that of the ground-truth lesions from 11.3 % to 9.3 %, without a loss in Dice (80.2 %); $|\Delta\mathrm{TLG}|$ fell from 14.5 % to 12.9 %. Larger factors reduced the change of the predicted masks further, to below 1 % from $\times 0.01$ on, but not that of the ground-truth lesions (9.0 to 9.8 %), while Dice decreased steadily, to 78.6 % at $\times 0.1$ and 78.0 % at $\times 1$. Compared with the network output before IO (dashed), IO with the PET losses thus canceled the increase that IO otherwise causes and ended slightly below it ($|\Delta\mathrm{MTV}|$ 9.3 % vs. 10.3 %, $|\Delta\mathrm{TLG}|$ 11.8 % vs. 12.5 % at $\times 0.1$), but not significantly (two-sided Wilcoxon signed-rank test, $p = 0.17$ and $p = 0.86$; lower for 14 and 13 of 26 patients), while gaining 3.2 Dice points. The factor chosen on the validation set ($\times 0.1$) thus cost 1.6 Dice points more than $\times 10^{-4}$ for the same volume change (9.3 %) and 1.1 points less uptake change (11.8 %). On the predicted masks, $|\Delta\mathrm{TLG}|$ fell from 10.2 % to 5.2 % at $\times 10^{-4}$ and to about 3 % from $\times 0.1$ on. The same gap between the labels IO optimizes and the ground truth appeared for the organs: without the PET losses, IO reached 92.9 % Dice on the predicted organ labels but 80.2 % on the ground-truth labels, starting from 79.0 % and 75.5 % for the network output.

In the diagnostic run with provided lesion masks (black diamonds), the measured changes fell from 9.3 % to 0.1 % for MTV and from 11.8 % to 4.5 % for TLG at the tuned factor; Dice rose from 78.6 % to 79.1 %. At $\times 1$, the changes were similar (0.2 % MTV, 4.0 % TLG; 78.6 % Dice), so a larger weight hardly reduced the remaining uptake change even with provided masks. The

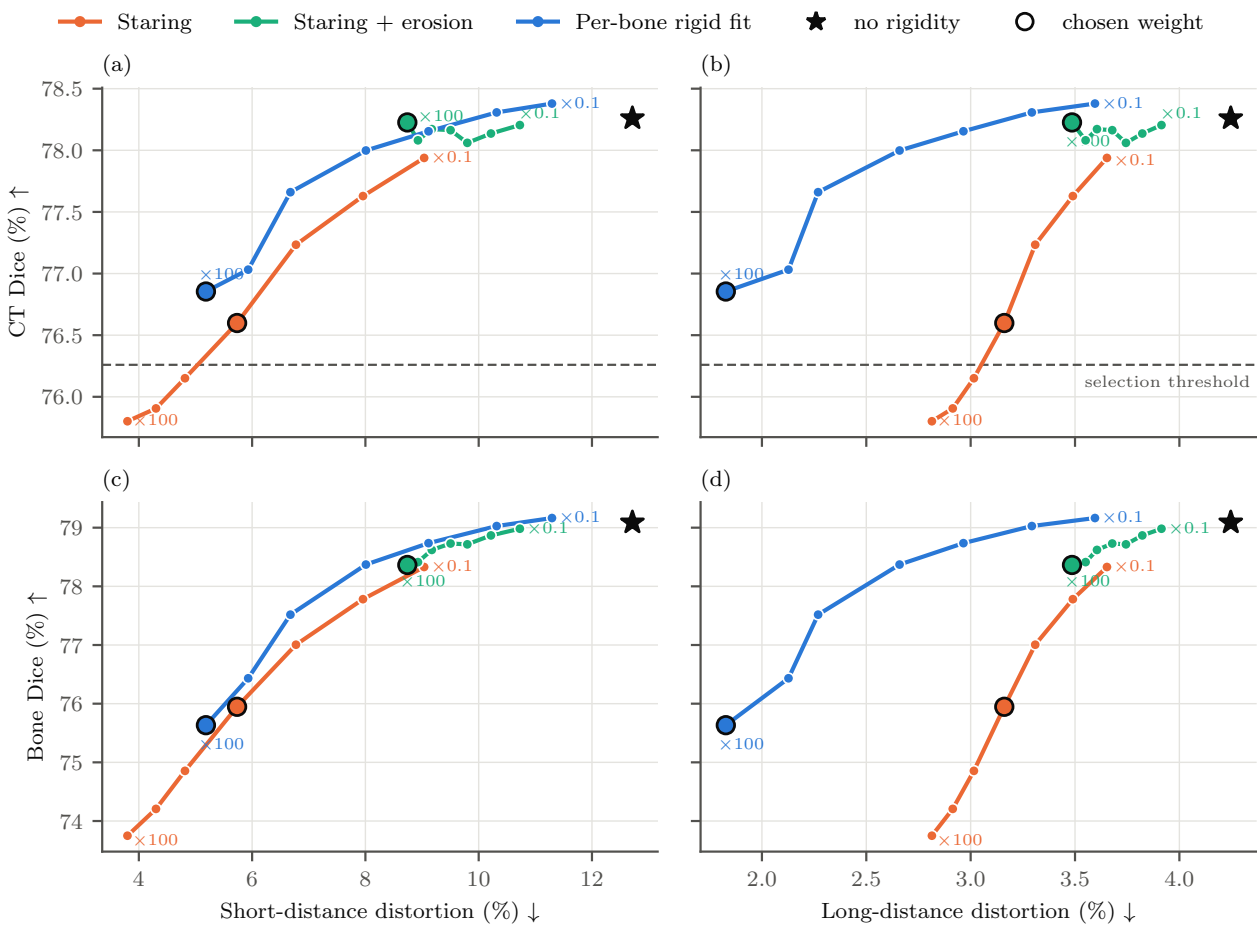


Fig 5 Rigidity weight sweep on the validation set: network trained with the PET losses, IO with the PET terms and one rigidity penalty at 0.1 to 100 times its calibrated reference weight (factors at the ends of each line). Top: CT Dice, bottom: bone Dice; left: distortion of short (5–15 mm), right: of long (≥15 mm) within-bone distances, i.e. the mean relative change of the distance between point pairs in the same bone. Star: variant without rigidity; dashed (a, b): selection threshold, two Dice points below it; outlined: weight chosen for the test set (the largest within the threshold, at most ×100). Points: patient means.

near-zero volume change is that of the scan total, which this run optimizes directly; the change of individual lesions remained (Sec. 4.3).

4.8 Lesion Coverage in Predicted Masks

To locate the remaining registration-induced change, we group all 1435 provided lesions, including the 108 smaller than five voxels, by how much each overlaps the mask used by IO (Fig. 9): 214 had no overlap (missed; 6 % of lesion volume), 155 had less than 50 % overlap (partly covered; 6 %), and 1066 had at least 50 % overlap (covered; 88 %).

PET IO reduced the change of covered lesions below that of the network output, from 9.1 % to 6.0 % for $|\Delta\mathrm{MTV}|$ and from 11.0 % to 8.4 % for $|\Delta\mathrm{TLG}|$ (IO without and with the PET losses at the tuned weights; network output 7.9 % and 9.2 %). Partly covered lesions changed as much with the PET losses as without them (12.5 % and 12.0 %), and missed lesions less (14.4 % and 17.0 %) but still more than at the network output (12.4 %). The loss has no direct target for a

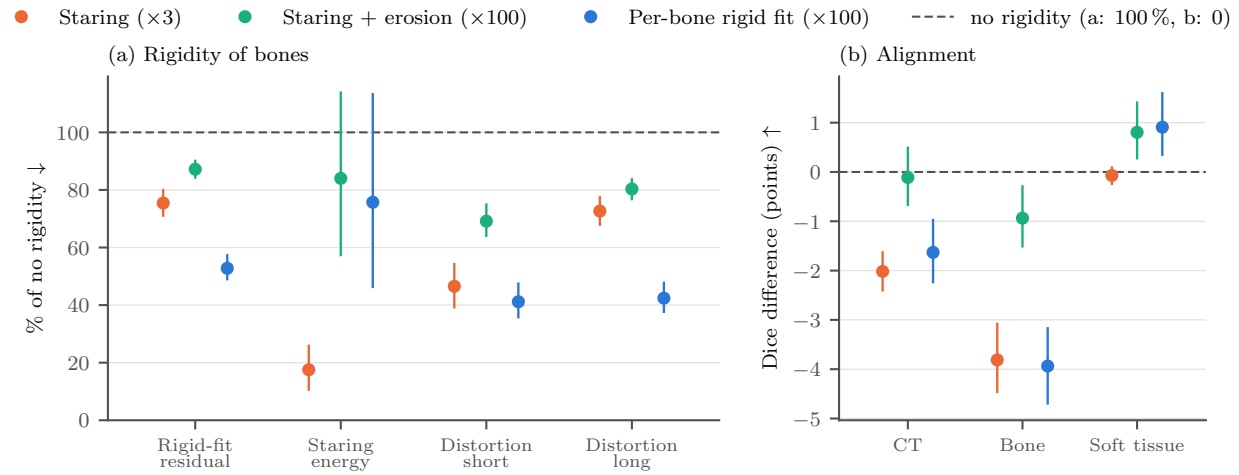


Fig 6 Rigidity penalties evaluated with all rigidity measures (test set, 28 patients): network trained with the PET losses, IO with the PET terms and one rigidity penalty at the weight chosen on the validation set. (a) Rigidity of bones relative to IO without rigidity (dashed, 100 %), per patient: residual of a rigid fit to each bone (the quantity the rigid-fit penalty minimizes), Staring energy, i.e. the Staring penalty (properness, orthonormality, and affinity) on the whole bone (the quantity the Staring penalty minimizes), and the distortion of short (5–15 mm) and long (≥15 mm) within-bone distances (minimized by neither). (b) Change of CT, bone, and soft-tissue Dice relative to IO without rigidity. Points: patient means; bars: 95 % patient-bootstrap confidence intervals.

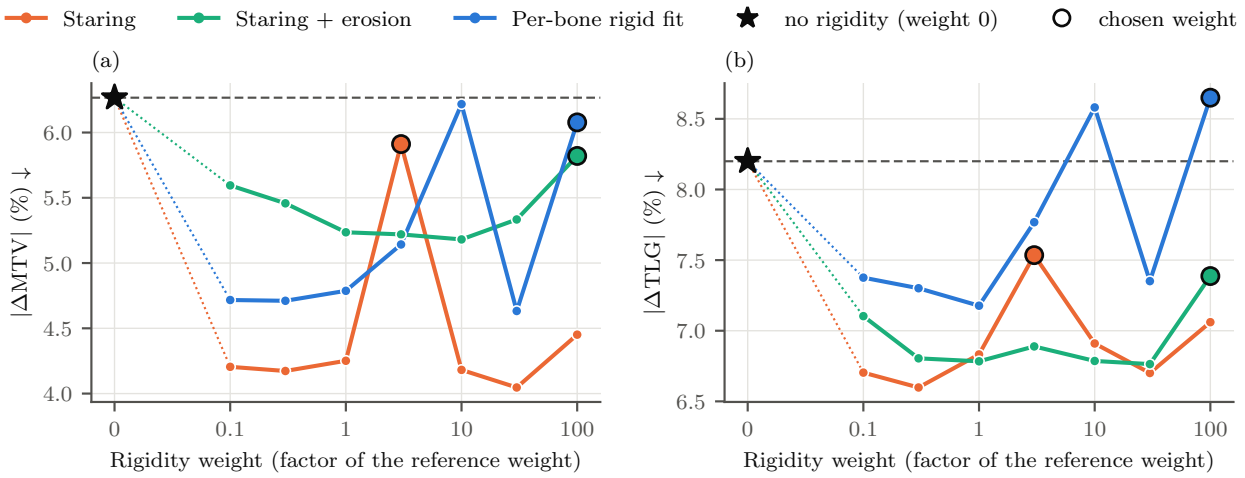


Fig 7 Effect of the rigidity weight on lesion preservation (validation set): network trained with the PET losses, IO with the PET terms and one rigidity penalty at 0.1 to 100 times its calibrated reference weight. (a) Lesion volume change, (b) uptake change. Star and dashed line: IO without rigidity (weight 0), dotted: link to the smallest weight; outlined: weight chosen for the test set. Points: patient means.

missed lesion and only a partial target for a partly covered lesion. After PET IO, missed lesions therefore contributed more than half of the remaining change: 5.2 of 9.3 points of $|\Delta\mathrm{MTV}|$ and 6.2 of 11.8 points of $|\Delta\mathrm{TLG}|$, although they make up only 6 % of the lesion volume. Covered lesions contributed 3.3 and 5.4 points. Scaling the PET weights from $\times 0.1$ to $\times 1$ changed none of these values by more than 0.3 points.

Without the PET losses, IO increased the change of every group over the network output, for covered lesions from 7.9 % to 9.1 % ($|\Delta\mathrm{MTV}|$) and from 9.2 % to 11.0 % ($|\Delta\mathrm{TLG}|$), and the

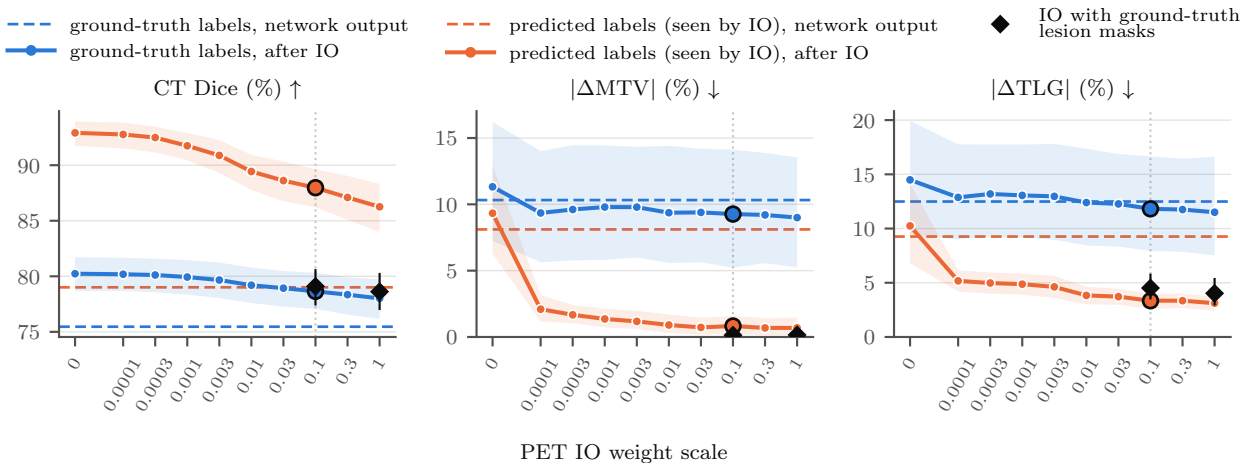


Fig 8 Effect of the PET loss weight in IO (test set): network trained without PET losses (T1) followed by IO with all PET weights scaled by the given factor (0: PET losses off; 1: the PET weights of network training). Each metric on the predicted labels that IO optimizes (orange; organs: TotalSegmentator, lesions: nnU-Net) and on the ground-truth labels used for evaluation (blue). Dice over all 28 patients, lesion volume and uptake change over the 26 patients with lesions in both label sets. Solid: after IO; dashed: network output before IO, the same for every factor. Outlined point and dotted line: operating point tuned on the validation set. Black diamonds: IO with the ground-truth instead of the predicted lesion masks (at $\times 0.1$ and $\times 1$), evaluated on the ground-truth labels. Points: patient means; bands: 95 % patient-bootstrap confidence intervals.

overall change from 10.3 % to 11.3 % and from 12.5 % to 14.5 %. This agrees with the per-lesion analysis (Sec. 4.3), in which IO without the losses increased the volume change of lesions. In the diagnostic run with provided lesion masks, the contribution of lesions that nnU-Net missed fell from 5.2 to 1.2 points of $|\Delta\mathrm{MTV}|$ and from 6.2 to 2.1 points of $|\Delta\mathrm{TLG}|$, and that of covered lesions from 3.3 to 0.9 and from 5.4 to 3.8 points. The near-zero overall change with provided masks does not mean that the lesions kept their volumes: if only the missed, partly covered, or covered lesions had changed, MTV would have changed by 1.2, 1.2, and 0.9 points, respectively. Some lesions therefore grew while others shrank, and their changes canceled in the scan total, which is what both the PET loss and the evaluation metric measure (see also Sec. 4.3).

5 Discussion

Compared with network output, IO with the PET-preservation and bone-rigidity losses improved alignment by about three Dice points without a detected increase in scan-level MTV or TLG change; IO without those losses increased both changes. This result supports loss-guided IO when predicted lesion masks and time for per-pair refinement are available. Applying the losses only during IO gave higher alignment accuracy than using them in both training and IO, although the paired intervals do not establish equivalent PET preservation. The remaining scan-level change arose largely from lesions missed by the predicted masks, while individual lesions still changed substantially even with provided masks. Bone-rigidity penalties addressed a separate anatomical aim: preserving within-bone geometry.

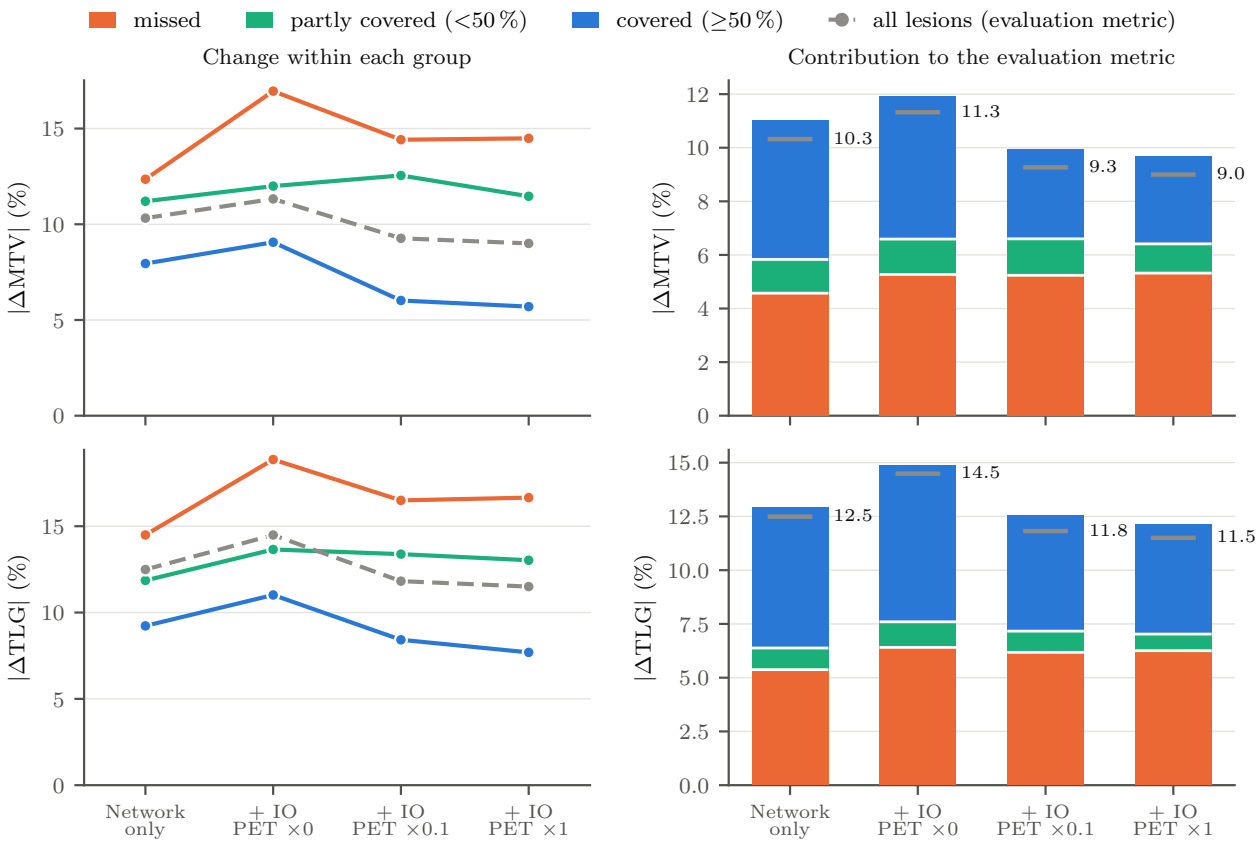


Fig 9 Lesion volume and uptake change of all 1435 lesions in 26 test patients (30 pairs), including components smaller than five voxels, grouped by how much of each lesion the predicted lesion mask seen by IO covers: missed (214 lesions, 6 % of volume), partly covered (155 lesions, 6 % of volume), covered (1066 lesions, 88 % of volume). Network trained without PET losses; IO with the PET weights scaled by the given factor. Left: change within each group (per scan, the change of the summed volume or uptake of the group's lesions; mean over the patients with lesions in the group); gray dashed: all lesions, i.e. the evaluation metric. Right: contribution of each group to the evaluation metric (the change per scan if only the group's lesions changed; mean over patients); gray ticks: the evaluation metric, below the stacked total because changes of different lesions in a scan partly cancel.

5.1 Choosing Where to Apply the Losses

Which placement to choose depends on the time budget and on the availability of predicted organ and lesion masks for IO. If IO is affordable, about 2.5 minutes per pair on our GPU, the network can be trained for accuracy alone and the losses applied during refinement. Compared with training and IO with both loss groups, this gained 1.0 Dice points. The paired intervals still allow worse MTV or TLG preservation with IO-only placement, so the result supports an accuracy gain but leaves the preservation trade-off uncertain. Without IO, the network alone registers a pair in 11 s, and training with the losses reduced registration-induced change to lesion volume and uptake. The training losses are therefore not superseded by IO, but they serve a different setting, e.g. fast registration of many follow-up examinations.

One possible explanation for the IO-only result is how each stage locates lesions. During network training, lesion labels define the preservation losses, but the trained registration network receives only images at inference and must respond to lesion-related image cues without a mask.

IO instead uses an explicit mask from a separately trained lesion-segmentation model, which can direct preservation to the predicted lesions in each pair. This could explain why adding the preservation losses during training provides little further measured preservation once IO is applied. The comparison does not test what lesion information the registration network learns, and the segmentation model cannot by itself explain the Dice advantage of IO-only placement: both IO variants use the same predicted masks.

5.2 Limits of Lesion Preservation

On the predicted lesion masks that IO optimizes, the PET losses reduced volume change from 9.3 % to 0.8 %; on the provided masks, 9.3 % MTV and 11.8 % TLG change remained. Increasing the PET weights tenfold had little effect on those measured changes. Missed lesions were the largest measured contributor to the residual (5.2 of 9.3 MTV points and 6.2 of 11.8 TLG points), although they make up only 6 % of lesion volume; covered lesions still contributed 3.3 and 5.4 points. Their weight comes from patients with little tumor volume: the metric is relative per scan and averaged over patients, and in the third of patients with the smallest lesion volume, missed lesions made up 33 % of the volume (23 % and 3 % in the other thirds) and caused 72 % of the missed-lesion contribution. Motivated by the recall-biased lesion mask of Dey and Dalca,[31] we also combined our nnU-Net predictions with SUV-thresholded regions after excluding dilated masks of organs with physiological uptake. On the test scans, this exploratory mask detected more lesions than nnU-Net alone (87 % vs. 74 %), but its voxel-level precision fell from 0.61 to 0.17. PET IO with this mask did not preserve better: the MTV and TLG changes were 10.3 % and 12.5 %, compared with 9.3 % and 11.8 % with the nnU-Net mask (differences not significant), and the per-lesion median was unchanged (13.5 % vs. 13.7 %). The additional false-positive regions may divert the preservation constraints from the lesions being evaluated. The diagnostic run with provided lesion masks reduced the measured changes from 9.3 % to 0.1 % for MTV and from 11.8 % to 4.5 % for TLG, while improving Dice from 78.6 % to 79.1 %. This supports mask quality as a major limit of the scan-level change, but uptake change remains even with provided masks, and individual lesions still change (below); the intervention does not establish that segmentation explains all residual error.

Because IO meets its objective on the predicted masks already at small weights, the PET weight mainly trades accuracy. A factor of 10^{-4} of the training-derived weights gave most of the preservation without a loss in Dice, whereas the factor chosen by the tuning on the validation set ($\times 0.1$) cost 1.6 points more for the same volume change and 1.1 points less uptake change; the tuning grid contained no factor between 0 and $\times 0.1$. Even with the ground-truth masks, a ten times larger weight hardly improved preservation.

In IO, the Jacobian terms carried the preservation, whereas removing the volume term $\mathcal{L}_{\mathrm{MTV}}$ or the uptake term $\mathcal{L}_{\mathrm{TLG}}$ improved $|\Delta\mathrm{TLG}|$ by 1.1 and 2.0 points, respectively, and removing $\mathcal{L}_{\mathrm{MTV}}$ also gained 0.6 Dice points. Both terms act on the soft, linearly warped mask, whose change differs from the nearest-neighbor change that is evaluated; a surrogate closer to the evaluated quantity, or the Jacobian terms alone, would likely preserve more. We kept all four terms because they were chosen before the test set was evaluated.

Aggregate MTV and TLG changes hide how individual lesions are deformed. Small lesions changed most in every condition, partly because a change of one or two voxels is already a large relative change. Without the losses, IO increased the distortion of lesions, most of all of small

bone-associated lesions, which tended to grow. The rigidity loss acted mainly on bone-associated lesions, as intended, and the PET losses on both bone-associated and soft-tissue lesions; with both groups of losses, bone-associated lesions changed about as little as with the PET losses alone. IO with the losses kept the scan-level measurements of the network output while improving alignment, but did not reduce the change of individual lesions, not even with the ground-truth lesion masks. The PET losses constrain the total volume and uptake of a scan, which can be preserved while single lesions grow or shrink.

Per-lesion terms on the connected components of the predicted mask did not solve this: they reduced the per-lesion change on the validation set but not on the test set. They leave the small lesions, which change most, unconstrained (components below 20 voxels and beyond the eight largest), they act on the predicted components, which miss lesions and merge or split others, and they use the same soft, linearly warped masks as the scan-level terms, whose change differs from the evaluated one (Sec. 4.4). A per-lesion constraint would likely need a closer surrogate of the nearest-neighbor evaluation and constraints for small lesions.

5.3 Interpreting the Evaluation

The significance-based ranking of the challenge was decided mainly by accuracy: differences in alignment between methods are consistent across patients and reach significance, whereas differences in MTV and TLG vary strongly between patients and rarely do. On the validation set, the ranking would consequently have chosen our variant without any of the losses; we therefore selected our method with a rule that weighs preservation within a fixed Dice tolerance (Sec. 3.3). Preservation metrics should therefore be read together with anatomical alignment; neither higher CT Dice nor lower MTV/TLG change alone establishes which registration is more useful for a clinical task. The scan-level MTV and TLG changes are also an incomplete measure of preservation: they sum over all lesions of a scan, so a lesion that grows can cancel one that shrinks. With provided lesion masks, IO reduced the scan-level volume change to 0.1 %, although single lesions still changed by a median of 13.5 %; a near-zero scan-level change therefore does not show that each lesion was preserved. Per-lesion changes, which do not cancel between lesions, should complement the scan-level metrics, in particular for tasks that follow individual lesions, such as lesion-level response assessment.

Both the measured alignment and preservation of IO depend in part on the labels it uses. On the predicted organ labels, IO raised Dice from 79.0 % to 88.0 %, on the ground-truth labels only from 75.5 % to 78.6 %; for lesions, it preserved the predicted masks almost perfectly but not the missed lesions. Within this segmentation-based evaluation, the diagnostic lesion-mask run shows that better lesion masks can improve the reported PET metrics; organ-mask quality may also affect the measured alignment. Outside this setting, label overlap and the change of MTV and TLG are proxies: what matters clinically is whether registration supports downstream tasks such as lesion tracking, response assessment, or dose accumulation. Future work should therefore evaluate registration on such tasks and clinical endpoints rather than optimize further for segmentation-based metrics.

The baseline comparison also illustrates the role of deformation regularity. NiftyReg exceeded the folding threshold on 5 of 38 validation pairs and 8 of 38 test pairs (21 %), with up to 0.37 % NDV. The ranking selected its least-regularized setting because accuracy separated the settings more than folding; a setting chosen on the validation set did not transfer to new patients. Our

variants exceeded the 0.005 % NDV threshold on at most two test pairs, with at most 0.026 % NDV.

5.4 Bone Rigidity

Finite-difference rigidity penalties face a trade-off at bone boundaries: their stencils read across the boundary (21 % of the reads), so they also stiffen the surrounding tissue, and eroding the bones to avoid this leaves most of the skeleton unconstrained at this resolution (35.5 % of bone voxels kept, ribs 2.4 %). The per-bone rigid fit avoids both, because it compares each bone's displacement with its best rigid motion and needs no derivatives. It preserved within-bone distances best, although it does not optimize them, and kept soft-tissue alignment, whereas Staring with erosion constrained the bones least. The cost of rigid bones is lower bone Dice, possibly because a rigid bone cannot follow differences between the automatic bone labels of the two examinations as closely as a freely deforming one. Whether this lower overlap reflects worse alignment or an inability of a rigid bone to follow inconsistencies between the automatic bone labels cannot be decided from label overlap alone. The lower within-bone distance distortion shows better preservation of bone shape, but it does not determine whether the bone is placed correctly.

Although about two thirds of the lesion voxels lie on bone labels (63 % in the training and 67 % in the test scans), adding a rigidity penalty on top of the PET losses did not measurably improve lesion volume or uptake preservation in this comparison. Rigidity alone did improve particular PET endpoints in Fig. 2: it reduced network-output $|\Delta\mathrm{MTV}|$ and IO $|\Delta\mathrm{TLG}|$ relative to the corresponding unconstrained variants. The absence of a detected additional benefit on top of the PET losses has three likely explanations. First, the PET losses already constrain the predicted lesions. Second, the bones were not made rigid at the scale of the lesions: the remaining rigid-fit residual of 1.4 to 2.3 mm is about half a voxel, enough to move the boundary voxels of lesions that span only a few voxels, whereas an exactly affine transformation changed bone-associated lesions by a median of only 1.0 %. Third, the scan-level metrics are dominated by missed lesions, by cancellation between lesions, and by a few patients with large changes (Sec. 4.8). The per-bone rigid fit is therefore useful when preserving internal bone geometry is an objective, with a measured cost in bone Dice. The present measures do not establish whether it improves bone placement, and a further PET-preservation benefit may require bones that are rigid to within a fraction of a voxel.

5.5 Limitations

The study uses a single cohort with 28 test patients, so small differences, in particular in MTV and TLG, could not be resolved, and generalization to other scanners and tracers is untested. The images were resampled to about $2.7\times2.7\times3.3$ mm, at which many lesions consist of few voxels and small lesions are dominated by discretization. The organ labels used for evaluation are themselves automatic (TotalSegmentator), and the lesion labels used by IO are predictions with a Dice of about 0.5, which, as shown above, limits preservation in IO. All patients, including the test patients, were part of the challenge data with which the earlier version of the method was developed; the test patients were not used for any decision in the present study. We studied one network architecture, and we measured the change caused by registration, not its effect on clinical decisions such as response assessment.

Table 4 Effect of the PET and rigidity losses and of where they act, on the test set (CT Dice: 28 patients, 38 pairs; PET: 26 patients, 30 pairs); values of Fig. 2. Left: network trained with the losses, without IO; middle: the same network followed by IO with the same losses; right: the network trained without losses followed by IO with the losses. Mean±SD over patients. †/‡: significantly better/worse than without losses in the same block (for the right block: the network without losses with IO); *: significantly different from the same losses in training and IO (two-sided Wilcoxon signed-rank tests on patient values, Holm-corrected over three comparisons within each column, $p < 0.05$).

	Losses only in training (no IO)			Losses in training and IO			Losses only in IO		
Losses	Dice (%) ↑	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓	Dice (%) ↑	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓	Dice (%) ↑	|ΔMTV| (%) ↓	|ΔTLG| (%) ↓
None	75.5±4.7	10.3±9.5	12.5±11.1	80.2±4.2	11.3±11.9	14.5±14.1	–	–	–
PET	73.6±4.6‡	8.2±9.1†	9.8±10.1	77.7±4.8‡	8.2±10.1	10.3±10.2	78.6±4.4$^{\ddagger *}$	9.3±11.6	11.8±11.7
Rigidity	75.5±4.7	8.9±8.4†	11.0±10.3	79.6±4.2‡	9.2±10.2	10.6±13.4†	79.6±4.2‡	10.0±11.3	11.5±14.2†
PET + rigidity	73.6±4.6‡	8.0±9.0†	9.8±10.9	77.3±4.9‡	7.9±9.7	10.3±10.4	78.3±4.4$^{\ddagger *}$	8.7±10.7	11.3±11.2

6 Conclusion

With predicted lesion masks, instance optimization with PET-preservation and bone-rigidity losses improved anatomical alignment without a detected increase in scan-level MTV or TLG change relative to network output; instance optimization without those losses increased both changes. Applied only during instance optimization, the losses gave higher alignment accuracy than when they were also used in training; the present test set did not resolve whether preservation differs between these placements. Without instance optimization, training with the losses reduced the change of lesion measurements at a cost in alignment, which suits fast registration of many examinations. Missed lesions accounted for much of the residual scan-level change in the mask-coverage analysis. Even with provided masks, substantial changes to individual lesions remained despite a near-zero scan-level MTV change. Better lesion masks may reduce scan-level error, while per-lesion evaluation is needed to reveal changes that whole-scan metrics hide. For bones, a per-bone rigid fit preserved within-bone distances best at a cost in bone Dice. Adding rigidity on top of the PET losses did not produce a detected further reduction in lesion-measurement change, although rigidity alone improved particular PET endpoints relative to the unconstrained variants.

7 Appendix A: Effect of the Losses

Table 4 gives the values and significance tests behind Fig. 2.

8 Appendix B: Per-Lesion Uptake Change

9 Appendix C: Tuning of Instance Optimization

Table 5 lists all candidates of the two IO tuning stages on the validation set (Sec. 3.3) and the selected settings.

10 Appendix D: Signed Change of Lesion Volume and Uptake

The main results report the absolute change of MTV and TLG. Table 6 gives the signed change, which shows whether a method systematically enlarges or shrinks the lesions. Every method enlarged the lesions on average (mean ΔMTV $+1.1$ to $+8.2\,\%$, ΔTLG $+2.8$ to $+9.6\,\%$), including the affine pre-registration ($+2.9\,\%$ and $+2.8\,\%$); the spread over patients was larger than these means for every deformable method.

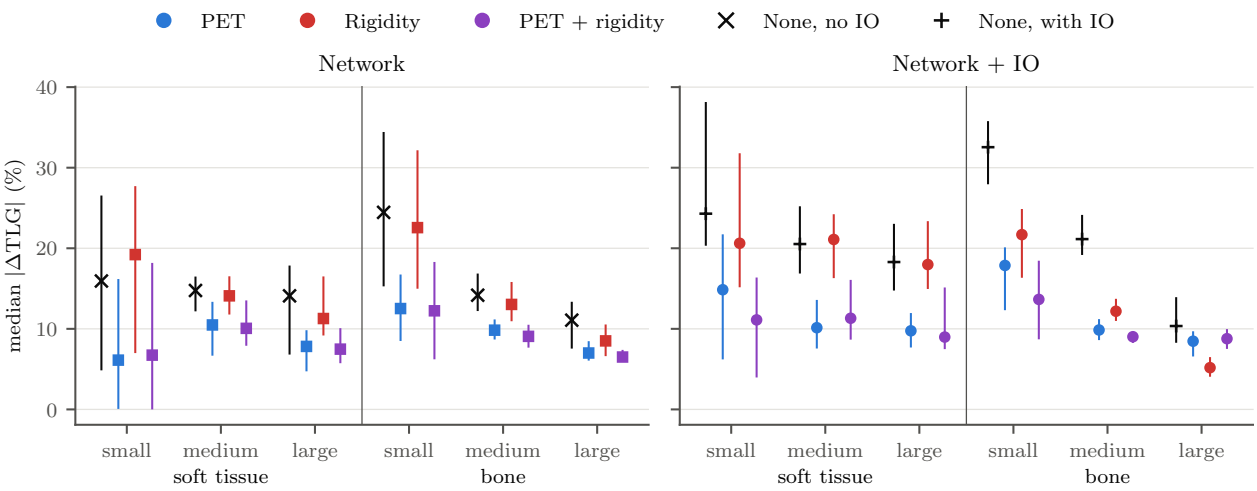


Fig 10 Per-lesion uptake change on the test set: median $|\Delta\mathrm{TLG}|$ per lesion, otherwise as Fig. 3.

11 Appendix E: Lesion Coverage in Predicted Masks

Tables 7 and 8 give the numbers behind Fig. 9.

12 Appendix F: Rigidity Penalties

Table 9 lists all measures of the rigidity comparison in absolute values, with tests against IO without rigidity, including the three terms of the Staring energy, HD95, and folding in and around the bones.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code, Data, and Materials Availability

All experiments use the publicly available training data of the Learn2Reg 2026 PSMAReg challenge,[3,4] which is adapted from the PSMA-PET-CT-Lesions dataset in The Cancer Imaging Archive[5] (DOI: 10.7937/r7ep-3x37). The private test data of the challenge were not used. The patient-level split into training, validation, and test sets is provided with the code. [TODO: Code repository link (training, IO, evaluation, split, and trained weights).]

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Table 5 IO tuning on the validation set (network trained with the PET and rigidity losses). Stage 1: learning rate / iterations; stage 2: common scale of the four PET weights relative to the training-derived values. Within each stage, the chosen setting (bold) has the lowest $|\Delta\mathrm{MTV}| + |\Delta\mathrm{TLG}|$ among the candidates within two Dice points of the most accurate one ("Within"). Candidates sorted by Dice; patient means. The rigidity weight was fixed at 0.25 (Sec. 3.3).

Candidate	Dice (%)	HD95 (mm)	$\lvert\Delta\mathrm{MTV}\rvert$ (%)	$\lvert\Delta\mathrm{TLG}\rvert$ (%)	Sum (%)	NDV (ppm)	Within
Stage 1: learning rate / iterations							
0.1 / 20	78.7	6.58	10.0	9.8	19.8	15	yes
0.1 / 40	78.6	6.58	10.9	10.6	21.5	13	yes
0.1 / 10	78.1	6.69	7.6	8.4	16.0	17	yes
0.05 / 40	78.0	6.78	6.9	8.1	15.0	29	yes
0.05 / 20	77.8	6.81	6.7	8.0	14.7	29	yes
0.05 / 10	77.0	6.84	5.3	7.0	12.2	34	yes
0.025 / 40	77.0	6.89	4.6	6.2	10.7	12	yes
0.025 / 20	76.7	6.93	4.6	6.3	10.9	12	yes
0.0125 / 40	76.1	6.99	3.9	5.7	9.6	2	no
0.025 / 10	75.7	7.14	4.3	5.5	9.8	18	no
0.0125 / 20	75.6	7.16	3.6	5.1	8.7	2	no
0.0125 / 10	75.3	7.17	3.2	4.6	7.8	3	no
Stage 2: PET weight scale (learning rate 0.025, 40 iterations)							
$\times 0$	79.5	6.61	10.1	11.5	21.6	0	yes
$\times$**0.1**	77.9	6.77	5.6	7.8	13.4	7	yes
$\times 0.25$	77.1	6.86	5.1	6.8	11.9	12	no
$\times 0.5$	77.1	6.87	4.9	6.6	11.5	12	no
$\times 1$	77.1	6.87	5.0	6.7	11.6	12	no
$\times 2$	77.0	6.89	4.9	6.6	11.5	13	no
$\times 4$	77.0	6.89	4.6	6.2	10.8	12	no

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References

1 Wolfgang P. Fendler, Matthias Eiber, Mohsen Beheshti … Ken Herrmann, "68Ga-PSMA PET/CT: Joint EANM and SNMMI procedure guideline for prostate cancer imaging: version 1.0," *European Journal of Nuclear Medicine and Molecular Imaging* **44**, 1014–1024 (2017).

2 Learn2Reg Consortium, "Learn2Reg 2026 – PSMAReg: Registration of pre-therapy and follow-up whole-body PSMA PET/CT scans." Codabench challenge (2026). https://www.codabench.org/competitions/15724/.

3 Lasse Hansen, Wiebke Heyer, Christoph Großbröhmer … Mattias P. Heinrich, "Learn2reg 2024: New benchmark datasets driving progress on new challenges," *Machine Learning for Biomedical Imaging* **3**, 775–791 (2025).

4 Alessa Hering, Lasse Hansen, Tony C. W. Mok … Mattias P. Heinrich, "Learn2reg: Comprehensive multi-task medical image registration challenge, dataset and evaluation in the era of deep learning," *IEEE Transactions on Medical Imaging* **42**(3), 697–712 (2023).

Table 6 Signed change of lesion volume and uptake caused by registration (test set, 26 patients with lesions): positive values mean that the warped lesions are larger or take up more. Methods as in Table 3 (Training / IO: losses used; PET + rig.: PET-preservation and rigidity losses). Mean±SD and median over patients.

			ΔMTV (%)		ΔTLG (%)	
Method	Training	IO	Mean±SD	Median	Mean±SD	Median
Affine	–	–	$+2.9 \pm 5.8$	$+1.3$	$+2.8 \pm 5.7$	$+1.7$
NiftyReg	–	–	$+4.6 \pm 14.7$	$+0.5$	$+6.3 \pm 18.7$	$+0.3$
ConvexAdam	–	–	$+8.2 \pm 12.8$	$+5.8$	$+9.6 \pm 17.2$	$+5.7$
Affine + IO	–	PET + rig.	$+3.5 \pm 5.8$	$+3.8$	$+5.7 \pm 5.6$	$+6.7$
Ours	–	–	$+5.3 \pm 13.1$	$+4.5$	$+7.9 \pm 14.9$	$+7.0$
Ours	PET + rig.	–	$+5.1 \pm 11.0$	$+3.4$	$+7.6 \pm 12.5$	$+5.2$
Ours + IO	–	basic	$+1.1 \pm 16.5$	$+2.1$	$+4.0 \pm 20.0$	$+3.6$
Ours + IO	PET + rig.	PET + rig.	$+4.2 \pm 11.8$	$+3.0$	$+7.3 \pm 12.8$	$+6.4$
Ours + IO	–	PET + rig.	$+3.2 \pm 13.5$	$+3.1$	$+6.2 \pm 14.8$	$+6.2$

Table 7 Volume and uptake change of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers (test set; network trained without PET losses, IO with the PET weights scaled by the given factor). Per scan, the change of the summed volume or uptake of the group's lesions relative to the unwarped moving scan, in %; mean over the patients with lesions in the group.

	Missed 214 lesions, 6 % of volume		Partly (<50 %) 155 lesions, 6 % of volume		Covered (≥50 %) 1066 lesions, 88 % of volume	
	|ΔMTV|	|ΔTLG|	|ΔMTV|	|ΔTLG|	|ΔMTV|	|ΔTLG|
Network only	12.4	14.5	11.2	11.9	7.9	9.2
+ IO, PET ×0	17.0	18.9	12.0	13.7	9.1	11.0
+ IO, PET ×0.1 (tuned)	14.4	16.5	12.5	13.4	6.0	8.4
+ IO, PET ×1	14.5	16.7	11.5	13.0	5.7	7.7

5 Katharina Jeblick, Stefan O. Schönberg, and others, "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions (PSMA-PET-CT-Lesions)." The Cancer Imaging Archive (2024). Version 1.

6 Tom Vercauteren, Xavier Pennec, Aymeric Perchant ... Nicholas Ayache, "Diffeomorphic demons: Efficient non-parametric image registration," *NeuroImage* **45**(1, Supplement 1), S61–S72 (2009).

7 B.B. Avants, C.L. Epstein, M. Grossman ... J.C. Gee, "Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain," *Medical Image Analysis* **12**(1), 26–41 (2008). Special Issue on The Third International Workshop on Biomedical Image Registration - WBIR 2006.

8 D. Rueckert, L.I. Sonoda, C. Hayes ... D.J. Hawkes, "Nonrigid registration using free-form deformations: application to breast mr images," *IEEE Transactions on Medical Imaging* **18**(8), 712–721 (1999).

9 Guha Balakrishnan, Amy Zhao, Mert R. Sabuncu ... Adrian V. Dalca, "An unsupervised learning model for deformable medical image registration," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, (2018).

Table 8 Contribution of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers, to the volume and uptake change of the evaluation (test set; network trained without PET losses, IO with the PET weights scaled by the given factor). Contribution of a group: the change per scan if only the group's lesions changed and all other lesions kept their volume and uptake, in %; mean over patients. All: the evaluation metric. Since changes of different lesions in a scan partly cancel, the contributions add up to slightly more.

	Missed 214 lesions, 6 % of volume		Partly (<50 %) 155 lesions, 6 % of volume		Covered (≥50 %) 1066 lesions, 88 % of volume		All lesions 1435 lesions	
	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$	$\lvert\Delta\text{MTV}\rvert$	$\lvert\Delta\text{TLG}\rvert$
Network only	4.6	5.4	1.3	1.0	5.2	6.5	10.3	12.5
+ IO, PET ×0	5.3	6.4	1.3	1.2	5.3	7.3	11.3	14.5
+ IO, PET ×0.1 (tuned)	5.2	6.2	1.4	1.0	3.3	5.4	9.3	11.8
+ IO, PET ×1	5.3	6.3	1.1	0.8	3.3	5.1	9.0	11.5

10 Adrian V. Dalca, Guha Balakrishnan, John Guttag … Mert R. Sabuncu, "Unsupervised learning for fast probabilistic diffeomorphic registration," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, Alejandro F. Frangi, Julia A. Schnabel, Christos Davatzikos … Gabor Fichtinger, Eds., 729–738, Springer International Publishing, (Cham) (2018).

11 Junyu Chen, Eric C. Frey, Yufan He … Yong Du, "Transmorph: Transformer for unsupervised medical image registration," *Medical Image Analysis* **82**, 102615 (2022).

12 Tony C. W. Mok and Albert C. S. Chung, "Large deformation diffeomorphic image registration with laplacian pyramid networks," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, Anne L. Martel, Purang Abolmaesumi, Danail Stoyanov … Leo Joskowicz, Eds., 211–221, Springer International Publishing, (Cham) (2020).

13 Fryderyk Kögl, Anna Reithmeir, Daniel M. Lang … Julia A. Schnabel, "PET quantification-preserving registration for longitudinal whole-body PSMA PET/CT," (2026). arXiv:2610.21451.

14 Jakob Wasserthal, Hanns-Christian Breit, Manfred T. Meyer … Martin Segeroth, "Totalsegmentator: Robust segmentation of 104 anatomic structures in CT images," *Radiology: Artificial Intelligence* **5**(5), e230024 (2023).

15 Katharina Jeblick, Balthasar Schachtner, Andreas Mittermeier … others, "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions," *Scientific Data* **13**, 1023 (2026).

16 Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl … Klaus H. Maier-Hein, "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods* **18**, 203–211 (2021).

17 Maximilian Rokuss, Balint Kovacs, Yannick Kirchhoff … Fabian Isensee, "From fdg to psma: A hitchhiker's guide to multitracer, multicenter lesion segmentation in pet/ct imaging," *ArXiv* (2024).

18 Hanna Siebert, Lasse Hansen, and Mattias P. Heinrich, "Fast 3d registration with accurate optimisation and little learning for learn2reg 2021," in *Biomedical Image Registration, Domain Generalisation and Out-of-Distribution Analysis*, Marc Aubreville, David Zimmerer, and Mattias Heinrich, Eds., 174–179, Springer International Publishing, (Cham) (2022).

Table 9 All measures of the rigidity comparison (test set, 28 patients; network trained with the PET losses, IO with the PET terms and one rigidity penalty at the weight chosen on the validation set). Rigidity of bones: residual of a rigid fit to each bone (minimized by the rigid-fit penalty), Staring energy (minimized by the Staring penalty), and distortion of short (5–15 mm) and long (≥15 mm) within-bone distances (minimized by neither). HD95 over bones and soft-tissue organs; near-bone soft tissue: soft-tissue organs within three voxels of a bone; Staring energy: sum of its properness, orthonormality, and affinity terms on the whole bone; NDV in bone band: in a three-voxel band around the bones. Mean±SD over patients; †/‡: significantly better/worse than without rigidity (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over the three penalties per row, $p < 0.05$).

Measure	No rigidity	Staring (×3)	Staring + erosion (×100)	Per-bone rigid fit (×100)
Alignment				
CT Dice (%)	77.7±4.8	75.7±4.3‡	77.6±4.6	76.1±4.6‡
CT HD95 (mm)	6.96±2.35	7.03±2.35‡	6.90±2.34	7.03±2.35
Bone Dice (%)	78.4±6.4	74.6±5.7‡	77.5±6.2‡	74.5±6.2‡
Bone HD95 (mm)	7.53±3.38	7.68±3.40‡	7.54±3.36	7.79±3.41‡
Soft-tissue Dice (%)	79.9±3.9	79.8±3.9	80.7±3.9†	80.8±4.0†
Soft-tissue HD95 (mm)	6.31±2.94	6.29±2.95	6.18±2.98†	6.18±3.01†
Near-bone soft-tissue Dice (%)	74.1±4.3	73.7±4.2‡	74.9±4.2†	74.8±4.2
Rigidity of bones				
Rigid-fit residual (mm)	2.67±0.52	2.00±0.48†	2.31±0.47†	1.39±0.40†
Staring energy	3.517±2.110	0.272±0.090†	1.803±0.650†	1.419±0.473†
Staring properness	0.182±0.100	0.022±0.009†	0.097±0.033†	0.093±0.027†
Staring orthonormality	1.431±0.840	0.160±0.066†	0.765±0.289†	0.733±0.261†
Staring affinity	1.903±1.180	0.090±0.040†	0.942±0.339†	0.594±0.205†
Distortion short (%)	13.41±3.62	5.53±1.04†	8.76±1.04†	5.01±1.13†
Distortion long (%)	4.44±0.83	3.17±0.68†	3.53±0.63†	1.83±0.62†
Regularity				
NDV (ppm)	7.9±29.2	7.9±29.3†	0.7±0.6	0.6±0.6†
NDV in bone (ppm)	6.5±4.6	0.3±1.2†	16.5±23.6‡	9.6±23.7
NDV in bone band (ppm)	1.9±1.7	0.7±0.9†	3.7±4.0‡	3.6±4.6‡
Preservation				
$\lvert\Delta\mathrm{MTV}\rvert$ (%)	8.2±10.1	7.1±8.7	7.9±10.0	8.0±8.9
$\lvert\Delta\mathrm{TLG}\rvert$ (%)	10.3±10.2	9.2±9.4	9.6±10.2	10.5±9.4

19 Brian B. Avants, Nicholas J. Tustison, Gang Song … James C. Gee, "A reproducible evaluation of ants similarity metric performance in brain image registration," *NeuroImage* **54**(3), 2033–2044 (2011).

20 Vincent Arsigny, Olivier Commowick, Xavier Pennec … Nicholas Ayache, "A log-Euclidean framework for statistics on diffeomorphisms," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, *Lecture Notes in Computer Science* **4190**, 924–931, Springer (2006).

21 Diederik P. Kingma and Jimmy Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, (2015).

22 Wolfgang Kabsch, "A solution for the best rotation to relate two sets of vectors," *Foundations of Crystallography* **32**(5), 922–923 (1976).

23 S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(4), 376–380 (1991).

24 Yihao Liu, Junyu Chen, Shuwen Wei ... Jerry Prince, "On finite difference jacobian computation in deformable image registration," *International Journal of Computer Vision* **132**, 3678–3688 (2024).
25 Marius Staring, Stefan Klein, and Josien P. W. Pluim, "A rigidity penalty term for nonrigid registration," *Medical Physics* **34**(11), 4098–4108 (2007).
26 Frank Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin* **1**(6), 80–83 (1945).
27 Sture Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics* **6**(2), 65–70 (1979).
28 Bradley Efron and Robert J. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall, New York (1993).
29 Marc Modat, Gerard R. Ridgway, Zeike A. Taylor ... Sébastien Ourselin, "Fast free-form deformation using graphics processing units," *Computer Methods and Programs in Biomedicine* **98**(3), 278–284 (2010). HP-MICCAI 2008.
30 Marc Modat, David M. Cash, Pankaj Daga ... Sébastien Ourselin, "Global image registration using a symmetric block-matching approach," *Journal of Medical Imaging* **1**(2), 024003 (2014).
31 Neel Dey and Adrian V. Dalca, "Registration is a representation problem: Solving whole-body pet/ct registration," (2026).

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